

Notes on Eisenstein series and Whittaker functions

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Contents

1	Whittaker functions	1
1.1	Local Whittaker functions	1
1.2	Global Whittaker functions	3
1.3	Whittaker sheaves and Laumon sheaves	4
1.4	Fourier Transform and the construction of automorphic sheaf	8
2	Whittaker categories	12
2.1	Drinfeld compactification and Whittaker patterns	12
2.2	Geometric Casselman-Shalika	14
3	local/global Whittaker categories, revisited	17
3.1	Twisted $\mathcal{D}$ -modules and factorizations	17
3.2	local/global Whittaker category	20
3.3	global Whittaker category	22
3.4	local-to-global Comparison	26

1 Whittaker functions

1.1 Local Whittaker functions

Throughout this section, we let F be a local (archimedean or non-archimedean) field. Let G be a quasi-split reductive group with a Borel B and N its unipotent radical. Let $\psi : N(F) \rightarrow \mathbb{C}^\times$ be a character. We say that ψ is generic if the restriction $\psi|_{N_{\alpha_i}(F)}$ is non-trivial for every simple root α_i .

For (generic) character ψ , we define the space of Whittaker functions as the following:

$$\mathcal{W}(\psi) := \{W \in C^\infty(G(F)) \mid W(ng) = \psi(n)W(g), \forall n \in N(F)\}$$

Now if (π, V) is an admissible representation of $G(F)$, we say a functional (continuous) $\lambda : V \rightarrow \mathbb{C}$ is **Whittaker** if it satisfies: $\lambda(\pi(n)\varphi) = \psi(n)\lambda(\varphi), \forall n \in N(F), \varphi \in V$. If such a non-zero Whittaker functional exists, we say (π, V) is **ψ -generic**. The relation between (generic) representations and Whittaker functions is that we can consider the so called *Whittaker model* of π .

Namely for representation (π, V) , we consider a $G(F)$ -equivariant continuous map $\Lambda : V \rightarrow \mathcal{W}(\psi)$ (notice there is a natural left $G(F)$ -action on $\mathcal{W}(\psi)$) such that its composition with evaluation map $\text{ev}_I : \mathcal{W}(\psi) \rightarrow \mathbb{C}; W \mapsto W(I_{G(F)})$ gives a Whittaker functional, the image under Λ is called the **ψ -Whittaker model** of π , denoted by $\mathcal{W}(\pi, \psi)$. It is easy to check that if π is ψ -generic, then a choice of Whittaker functional λ gives rise to a Whittaker model by setting $\Lambda(\varphi)(g) = \lambda(\pi(g)\varphi)$. Moreover, we have the following:

Theorem 1.1 (Gelfand-Kazhdan, Shalika). *For a fixed generic character ψ , the space of Whittaker functional of an irreducible admissible representation π has at most one dimension.*

Thus as a corollary of Theorem 1.1, we can deduce that for ψ generic and π generic, the Whittaker model exists and is unique upto scalar. We now proceed to consider the Whittaker models for certain representations.

Iwasawa decomposition gives us $G(F) = \coprod_{\check{\mu} \in X_*(T)} N(F)\check{\mu}(\varpi)K$, where ϖ is the uniformizer of F and K a nice enough maximal compact subgroup. So now we eager to know the value of ψ -Whittaker models on $\check{\mu}(\varpi)$. Any irreducible unramified representation is equivalent to $J(\lambda)$, the unramified subquotient of $I(\lambda) = \text{Ind}_B^G(t \mapsto e^{\langle H_T(t), \lambda \rangle})$, where $\lambda \in X^*(T)_{\mathbb{C}}$ complexified character. Assume that $J(\lambda)$ is ψ -generic, we consider the space $\mathcal{W}(J(\lambda), \psi)^K$ of right K -invariant Whittaker model of $J(\lambda)$. We know by genericity that $\mathcal{W}(J(\lambda), \psi)$ is non-empty and $\mathcal{W}(J(\lambda), \psi)^K$ is one dimensional. The value of elements in $\mathcal{W}(J(\lambda), \psi)^K$ is determined:

Theorem 1.2 (Casselman-Shalika). *Let $W \in \mathcal{W}(J(\lambda), \psi)^K$ be a non-zero elements, then $W(I_{G(F)}) \neq 0$ and*

$$\frac{W(\check{\mu}(\varpi))}{W(I_{G(F)})} = \begin{cases} \delta_B^{\frac{1}{2}}(\check{\mu}(\varpi))\chi_{\check{\mu}}(q^{-\lambda}) & \text{when } \check{\mu} \text{ dominant coweight,} \\ 0 & \text{non-dominant coweight } \check{\mu}. \end{cases}$$

Here δ_B is a quasi-character occurs to balance Haar measure, $\chi_{\check{\mu}}$ is the character of the irreducible highest weight representation $V(\check{\mu})$ of $\check{G}$ and q denotes the size of residue.

Example 1.3. One can works out the explicit formulae for $G = \text{GL}_n$ easily according to Theorem 1.2. Any dominant coweight $\check{\mu}$ now is given by a sequence of number $(k_1, \dots, k_n) \in \mathbb{Z}^n$ and $\check{\mu}$ is dominant if and only if

$k_1 \geq k_2 \geq \dots \geq k_n$. In the case when $\check{\mu}$ is dominant, we normalize $W \in W(J(\lambda), \psi)^{\text{GL}_n(O_F)}$ so that it takes value 1 on $I_{\text{GL}_n(F)}$, then the value of W on $\check{\mu}(\varpi)$ is given by

$$W\left(\check{\mu}(\varpi) = \begin{pmatrix} \varpi^{k_1} & & \\ & \ddots & \\ & & \varpi^{k_n} \end{pmatrix}\right) = \delta_{B_n}^{\frac{1}{2}}(\check{\mu}(\varpi)) \mathbb{S}_{\check{\mu}}(q^{-\lambda_1}, \dots, q^{-\lambda_n}), \text{ where } \mathbb{S} \text{ is the Schur-Weyl duality.}$$

Remark 1.4. As mentioned above, there are bijections between semi-simple conjugacy classes of $\check{G}$ and irreducible unramified representations of $G(F)$ and characters of the Grothendieck ring $K_0(\text{Rep } \check{G})$. We can write W in the form W_γ to indicate it is the Whittaker model of the unramified representation associated with γ . This notation coincides with the one in [Fre+98], in which $W_\gamma(\check{\mu}(\varpi))$ is written as $q^{-\langle \check{\mu}, \rho \rangle} \text{Tr}(\gamma, V(\check{\mu}))$.

From the Remark 1.4, we observe also that Whittaker functions are closely related to $K_0(\text{Rep } \check{G})$ as the character ϕ_γ of $K_0(\text{Rep } \check{G})$ corresponding to γ is given by $[V] \mapsto \text{Tr}(\gamma, V)$. Also we recall the classical Satake isomorphism tells us that the spherical Hecke algebra $(\mathcal{H}_G = C_c^\infty(G(O_F) \backslash G(F) / G(O_F)), \star)$ is canonically isomorphic to $K_0(\text{Rep } \check{G})$ as an algebra. There are several choices of basis of $\mathcal{H}_G$, the natural one comes from the Cartan decomposition that $G(F) = \coprod_{\check{\mu} \in X_*(T)^+} G(O_F) \check{\mu}(\varpi) G(O_F)$ and one take $c_{\check{\mu}}$ be the characteristic function supported on $G(O_F) \check{\mu}(\varpi) G(O_F)$. The another basis is given by Satake isomorphism, we let $A_{\check{\mu}}$ be the image of $[V(\check{\mu})]$, the irreducible representation of highest weight $\check{\mu}$ of $\check{G}$ under Satake.

In [Fre+98], a certain $\mathcal{H}_G$ -module $\mathbf{W}_\psi$ is considered, it contains functions in $C_c^\infty(G(F))^{G(O)}$ that satisfies

$$f(n_g) = \chi(n) f(g)$$

where $\chi : N(F) \rightarrow \mathbb{F}_q$ is defined as $\chi(n) = \sum_{i=1}^{\text{rk}(G)} \chi(\text{Res}(n_i))$, and $n_i \in N_{\alpha_i}$ corresponding to the simple root. $\mathcal{H}_G$ acts on $\mathbf{W}_\psi$ via convolution (one can check that it preserves the space $\mathbf{W}_\psi$). It is obvious to see from Iwasawa decomposition that $f \in \mathbf{W}_\psi$ supported only on union of $N(F) \check{\mu}(\varpi) G(O_F)$ where $\check{\mu}$ dominant. There is a basis $\{\phi_{\check{\mu}}\}$ of $\mathbf{W}_\psi$ so that $\phi_{\check{\mu}}$ is supported on $N(F) \check{\mu}(\varpi) G(O_F)$ and $\phi_{\check{\mu}}(\check{\mu}(\varpi)) = q^{-\langle \check{\mu}, \rho \rangle}$. It is also shown in [Fre+98] that $\mathcal{H}_G \rightarrow \mathbf{W}_\psi : A_{\check{\mu}} \mapsto \phi_{\check{\mu}}$ is an isomorphism of $\mathcal{H}_G$ -module. Moreover, the Whittaker function W_γ can be written as

$$W_\gamma = \sum_{\check{\mu} \in \check{\Lambda}^+} \phi_{\check{\mu}} \cdot \text{Tr}(\gamma, V(\check{\mu}))$$

Finally, there is an important property of Whittaker functions called Hecke eigen-property. Namely if we consider the Hecke algebra action on $\mathcal{W}(\gamma, \psi)^K$ via convolution, then W_γ is a Hecke eigenform with eigenvalue γ , i.e. $h \star W_\gamma = \phi_\gamma(h) W_\gamma$, where ϕ_γ is defined on $K_0(\text{Rep } \check{G})$, hence on $\mathcal{H}_G$ via Satake.

1.2 Global Whittaker functions

In this section F is a global field and representations we consider are those $\pi \subseteq L_{\text{cusp}}^2([G] = A_G(F) \backslash G(\mathbb{A}_F))$, i.e. cuspidal representations. Fix a character $\psi : N(F) \backslash N(\mathbb{A}_F) \rightarrow \mathbb{C}^\times$ that is generic (i.e. generic at each place), we can similarly formulate the space of ψ -Whittaker functions:

$$\mathcal{W}(\psi) = \{W \in C^\infty([G]) \mid W(ng) = \psi(n)W(g), \forall n \in N(\mathbb{A}_F)\}.$$

For any vector φ in cuspidal representation π , we can associate it with $\mathcal{W}(\psi) \ni W_\psi^\varphi : g \mapsto \int_{[N]} \varphi(ng) \bar{\psi}(n) dn$. Similar to the local case, we say π is ψ -**generic** if there is $\varphi \in \pi$ such that $W_\psi^\varphi \neq 0$. For (π, V) a generic cuspidal representation, note that we can write V as $V \cong (\hat{\otimes}_{v|\infty} V_v) \otimes (\hat{\otimes}_{v \nmid \infty}' V_v)$. Suppose that V_v can be regarded as space of functions that is contained $\{\varphi_v : G(F_v) \rightarrow \mathbb{C}^\times\}$, then to compute W_ψ^φ can be reduced to compute local Whittaker factors $W_{\psi_v}^{\varphi_v}$ (which is defined in the same way as W_ψ^φ). Notice that $\varphi \in V \subseteq L_{\text{cusp}}^2([G])$ usually is not the product of φ_v via the above isomorphism, however we have the following for Whittaker functions:

Proposition 1.5 (local-global compatibility). *Let (π, V) be as above, write $\mathcal{W}(\pi, \psi)$ be the space of global Whittaker functions (i.e. contains those W_ψ^φ), then we have the isomorphism*

$$\mathcal{W}(\pi, \psi) \cong (\hat{\otimes}_{v|\infty} \mathcal{W}(\pi_v, \psi_v)) \otimes \left(\hat{\otimes}_{v \nmid \infty}' \mathcal{W}(\pi_v, \psi_v) \right).$$

Moreover, there exists $c \neq 0$ such that $\mathcal{W}_\pi^\psi(g) = c \prod_v \mathcal{W}_{\pi_v}^{\psi_v}(g_v)$

Similar to the local case, we can also consider the Hecke algebra action on $\mathcal{W}(\pi, \psi)^{G(\mathbb{O}_F)}$ (at each place). From local-global compatibility we can deduce the Hecke eigen-property for global Whittaker functions.

One particular example is when $G = \text{GL}_n$, then every cuspidal representation is ψ -generic whenever ψ is generic. Moreover, in this case the Whittaker functions play the role of Fourier expansion.

Theorem 1.6. *Let $\varphi \in L_{\text{cusp}}^2([\text{GL}_n])$ be a smooth vector and assume ψ is generic, then we have*

$$\varphi(g) = \sum_{\gamma \in N_{n-1}(F) \backslash \text{GL}_{n-1}(F)} W_\psi^\varphi \left(\begin{pmatrix} \gamma & \\ & 1 \end{pmatrix} \cdot g \right)$$

where N_{n-1} denotes the unipotent subgroup of GL_{n-1} .

In particular, the Theorem 1.6 implies the following theorem by Shalika:

Theorem 1.7 (Multiplicity one Theorem for GL_n). *Any irreducible admissible representation occurs in $L_{\text{cusp}}^2([\text{GL}_n])$ at most once.*

The collection of $\begin{pmatrix} \gamma & \\ & 1 \end{pmatrix}$ that occurs in the Whittaker expansion is contained in so called mirabolic subgroup of GL_n . More precisely, we let $\text{Mir}_n := \left\{ \begin{pmatrix} \gamma & v \\ & 1 \end{pmatrix} \mid \gamma \in \text{GL}_{n-1} \right\} \subseteq \text{GL}_n$.

Proposition 1.8. *There is an $\mathrm{GL}_n(\mathbb{A}_F)$ -equivariant isomorphism*

$$\mathbf{Av} : \mathcal{W}(\psi) \xrightarrow{\cong} C_{\mathrm{cusp}}^\infty(\mathrm{Mir}_n(F) \backslash \mathrm{GL}_n(\mathbb{A}_F)), \quad W \mapsto \left(g \mapsto \sum_{\gamma \in (N_{n-1} \backslash \mathrm{GL}_{n-1}(F))} W \left(\begin{pmatrix} \gamma & \\ & 1 \end{pmatrix} g \right) \right)$$

Let us assume F is the function field of a curve (smooth, proper and connected), let $\sigma : \mathrm{Gal}_F \rightarrow \mathrm{GL}_n(\mathbb{C})$ (we do not distinguish $\mathbb{C}$ and $\bar{\mathbb{Q}}_\ell$ for simplicity) be an irreducible everywhere unramified representation. Then at each point $x \in X$, $\sigma(\mathrm{Forb}_x)$ can be regarded as a semisimple conjugacy classes in GL_n , write $W_{\sigma(\mathrm{Forb}_x),x}$ the corresponding local Whittaker function (with respect to a generic ψ). Define

$$W_\sigma : \mathrm{GL}_n(\mathbb{A}_F) \rightarrow \mathbb{C}, \quad g \mapsto \prod_{x \in |X|} W_{\sigma(\mathrm{Forb}_x),x}(g_x)$$

One can check that $W_\sigma \in \mathcal{W}(\psi)$, then apply $\mathbf{Av}$ we get $f'_\sigma = \mathbf{Av}(W_\sigma) \in C_{\mathrm{cusp}}^\infty(\mathrm{Mir}_n(F) \backslash \mathrm{GL}_n(\mathbb{A}_F))$. This is quite close to the expectation of (unramified) Langlands correspondence that such a unramified L-parameter σ should correspond to a function $f_\sigma : G(F) \backslash G(\mathbb{A}_F) / G(\mathbb{O}_F) \rightarrow \mathbb{C}$ that satisfies the Hecke eigen-property. The Hecke action is given as: for $\forall i = 1, \dots, n, \forall x \in |X|$, the Hecke operator acts as

$$T_x^i : f \mapsto \left(g \mapsto \int_{\mathrm{GL}_n(\mathcal{O}_x) \mathrm{diag}(\underbrace{\varpi_x, \dots, \varpi_x}_{i\text{-times}}, 1, \dots, 1) \mathrm{GL}_n(\mathcal{O}_x)} f(gh) dh \right), \quad \text{where } \varpi \text{ uniformizer in } \mathcal{O}_x \quad (1.2.1)$$

Precisely the Hecke eigen-property says f_σ should satisfies $T_x^i(f_\sigma) = \mathrm{Tr}(\Lambda^i \sigma(\mathrm{Frob}_x)) \cdot f_\sigma$. The right invariance is not a big issue, however the left invariance by Mirabolic subgroup could not easily lift to $\mathrm{GL}_n(F)$. The geometric construction thus comes to play.

1.3 Whittaker sheaves and Laumon sheaves

Throughout this section, we work over function field F of a curve over $\mathbf{k} = \mathbb{F}_q$ and the group G is taken to be GL_n . In [Fre+98], there is a twisted version of isomorphism in Proposition 1.8 we now explain: Let $\dot{G}$ be the ρ -shift of GL_n , elements of $\dot{G}(R)$ can be written as $\{(a_{ij}) \mid a_{ij} \in \Omega_R^{j-i}, \det \neq 0\}$, where R is taken to be F_x or F . Similarly, one can also define $\dot{T}, \dot{B}, \dot{N}$ etc. In addition, we set $\dot{T}(F_x)^+ := \dot{T}(F_x) \cap \mathrm{Diagn}(\mathcal{O}_x)$, where $\mathrm{Diagn}(\mathcal{O}_x)$ denotes diagonal matrices with entries in $\mathcal{O}_x$. We also have the corresponding $\dot{B}(F_x)^+ := \dot{T}(F_x)^+ \dot{N}(F_x)$. Also, let $\dot{P}(F_x)^+ = \left\{ \begin{pmatrix} a & b \\ & c \end{pmatrix} \mid a \in \dot{\mathrm{GL}}_{n-1}(F_x), c \in F_x^\times, |c| \leq 1 \right\} \subseteq \dot{P}_{n-1,n}(F_x)$. Finally, we consider three sets:

$$\begin{aligned} Q &= \dot{N}(F) \backslash \dot{B}(\mathbb{A}_F)^+ / \dot{B}(\mathbb{O}_F) \subseteq \dot{N}(F) \backslash \dot{G}(\mathbb{A}_F) / \dot{G}(\mathbb{O}_F) \\ M'_n &= \dot{\mathrm{Mir}}(F) \backslash \dot{P}(\mathbb{A}_F)^+ / \dot{P}(\mathbb{O}_F) \subseteq \dot{\mathrm{Mir}}(F) \backslash \dot{G}(\mathbb{A}_F) / \dot{G}(\mathbb{O}_F) \\ M_n &= \dot{G}(F) \backslash \dot{G}(\mathbb{A}_F) / \dot{G}(\mathbb{O}_F) \end{aligned} \quad (1.3.1)$$

And there are obvious projection maps $\mu : Q \rightarrow M'_n, p : M'_n \rightarrow M_n$, moreover we have the following bijections:

Proposition 1.9. *The set Q is in bijection with the isomorphism classes of triples*

$$\{(\mathcal{E}, F_\bullet, (s_i)) \mid \mathcal{E} \in \text{Bun}_n(\mathbf{k}); F_\bullet \text{ is a flag } 0 = F_n \subseteq F_{n-1} \subseteq \dots \subseteq F_0 = \mathcal{E}; s_i : \Omega^i = (\Omega^1)^{\otimes i} \rightarrow F_i/F_{i+1}\}$$

The set M'_n is in bijection with $\{(\mathcal{E}, s) \mid \mathcal{E} \in \text{Bun}_n(\mathbf{k}), s : \Omega^{n-1} \hookrightarrow \mathcal{E}\}$ and the last set M_n is in bijection with the set of isomorphism classes of rank n bundles on X . Via those bijections, the projection μ is forgetting the flag structure and maps (s_i) to s_{n-1} , the projection p is just the forgetful map.

Proof. Let J^0 denote the rank n vector bundle $\bigoplus_{i=0}^{n-1} \Omega^i$, we consider the data of triples $\{(\mathcal{E}, \beta_\eta, (\beta_x)_{x \in X})\}$, where $\mathcal{E} \in \text{Bun}_n(\mathbf{k})$, $\beta_\eta : J_\eta^0 \xrightarrow{\cong} \mathcal{E}_\eta$ and $\beta_x : J_{O_x}^0 \xrightarrow{\cong} \mathcal{E}_{O_x}$ are generic and local trivialization respectively. For any $x \in |X|$, we have an automorphism $\mu_{\beta, x}$ on $J_{F_x}^0$ given by $(\beta_x)_{F_x}^{-1} \circ (\beta_\eta)_{F_x}$. This provides an elements g_x in $\dot{G}(F_x)$ whose entries are given by $g_{x,ij} := \Omega_{F_x}^i \hookrightarrow J_{F_x}^0 \xrightarrow{\mu_{\beta, x}} J_{F_x}^0 \twoheadrightarrow \Omega_{F_x}^k$. Now for any $(\mathcal{E}, F_\bullet, (s_i))$ as in the Proposition, we can choose β_η, β_x so that whenever $R = F, O_x \bigoplus_{i=j}^{n-1} \Omega_R^i$ maps to $F_{j,R}$ and the induced map $\Omega^j \rightarrow F_j/F_{j+1}$ coincides with s_i . Such choice of β forces the corresponding g_x lies in $\dot{B}(F_x)^+$. The freedom of choice of β_η lies in $\dot{N}(F)$ while β_x lies in $\dot{B}(O_x)$. $\square$

From the last Proposition, we can see that the stacky version of M_n is nothing but Bun_n . So the question to find automorphic form (i.e. functions on M_n) is translated to find Hecke eigensheaves on Bun_n . Also notice that the functions W_σ and f'_σ we constructed in the previous section are determined by restriction to Q and M'_n respectively. The strategy in [FGV01a], roughly speaking, is to geometrize those double cosets in the above Proposition and study the sheaf analogs of W_σ and f'_σ .

We define stack Coh'_n so that $\forall S \in \text{Sch}_{\mathbf{k}}, S \rightarrow \text{Coh}'_n$ is groupoid consists of data $(\mathcal{E}, s)$, where $\mathcal{E}$ is a coherent sheaf on X_S of generic rank n and $s : \Omega_X^{n-1} \boxtimes \mathcal{O}_S \hookrightarrow \mathcal{E}$ is an embedding. Similarly the stack Bun'_n is a substack of Coh'_n whose S -points classifies data $(\mathcal{E}, s)$ while $\mathcal{E}$ is a vector bundle over X_S and $s : \Omega_X^{n-1} \boxtimes \mathcal{O}_S \hookrightarrow \mathcal{E}$ is still an embedding. We can observe that the underlying equivalent classes of Bun'_n coincides with M'_n .

Also we set three stacks related to set Q above, for $S \in \text{Sch}_{\mathbf{k}}$, they classify the following groupoids:

$$\begin{aligned} \tilde{Q}(S) &= \left\langle (\mathcal{E}', (\mathcal{E}_i), (\tilde{s}_i), \beta) \left| \begin{array}{l} \mathcal{E}' \in \text{Coh}_n(S), \mathcal{E}_i \in \text{Bun}_i(S), 0 = \mathcal{E}_0 \subseteq \mathcal{E}_1 \subseteq \dots \subseteq \mathcal{E}_n \text{ full flag of subbundle,} \\ \beta : \mathcal{E}_n \hookrightarrow \mathcal{E}' \text{ embedding of } \mathcal{O}_{X_S}\text{-module, } \tilde{s}_i : \Omega^{n-i} \boxtimes \mathcal{O}_S \xrightarrow{\cong} \mathcal{E}_i/\mathcal{E}_{i-1} \end{array} \right. \right\rangle \\ \overline{Q}(S) &= \left\langle (\mathcal{E}, s_i) \left| \begin{array}{l} \mathcal{E} \in \text{Bun}_n(S), s_i : \Omega^{\sum_{j=1}^i (n-j)} \boxtimes \mathcal{O}_S \hookrightarrow \Lambda^i \mathcal{E} \text{ embedding of } \mathcal{O}_{X_S}\text{-module} \\ s_i\text{'s satisfies Plucker relations} \end{array} \right. \right\rangle \\ Q(S) &= \left\langle (\mathcal{E}, s_i) \left| \begin{array}{l} \mathcal{E} \in \text{Bun}_n(S), s_i : \Omega^{\sum_{j=1}^i (n-j)} \boxtimes \mathcal{O}_S \hookrightarrow \Lambda^i \mathcal{E} \text{ maximal embeddings} \\ s_i\text{'s satisfies Plucker relations} \end{array} \right. \right\rangle \end{aligned} \quad (1.3.2)$$

Here by maximal embedding we mean a subbundle inclusion. Also when s_i are all maximal embeddings, they determines a full flag of subbundles of $\mathcal{E}$. In particular, this exhibits Q as an open substack of $\tilde{Q}$ that the first coordinate in $\tilde{Q}$ is replaced by $\mathcal{E}$, and the flag contained in $\mathcal{E}$ is determined by s_i 's.

To conclude, we have the following diagram of stacks, which is crucial for later construction:

$$\begin{array}{ccccc}
& \tilde{Q} & \xleftarrow{\tilde{j}} & Q & \xrightarrow{\bar{j}} & \bar{Q} \\
& \searrow \alpha & \downarrow \tilde{\nu} & & \downarrow \bar{\nu} & \\
\text{Coh}_0 & & \text{Coh}'_n & \xleftarrow{\quad} & \text{Bun}'_n & \\
& & \downarrow \rho_n^{\text{coh}} & & \downarrow \rho_n^{\text{bun}} & \\
& & \text{Coh}_n & \xleftarrow{\quad} & \text{Bun}_n &
\end{array} \tag{1.3.3}$$

where $\tilde{\nu} : (\mathcal{E}', (\mathcal{E}_i), (\tilde{s}_i), \beta) \mapsto (\mathcal{E}', \beta \circ \tilde{s}_1)$, $\bar{\nu} : (\mathcal{E}, (s_i)) \mapsto (\mathcal{E}, s_1)$ and the other two horizontal arrows are natural projection maps. The map $\alpha : \tilde{Q} \rightarrow \text{Coh}_0$ is given by $(\mathcal{E}', (\mathcal{E}_i), (\tilde{s}_i), \beta) \mapsto \mathcal{E}'/\text{Im } \beta$. Notice there is also an evaluation map $\text{ev} : \tilde{Q} \rightarrow \mathbb{G}_a$ given by the composition:

$$\text{ev} : \tilde{Q} \longrightarrow \prod_{i=1}^{n-1} \mathcal{E}xt^1(\Omega^{i-1}, \Omega^i) \longrightarrow \prod_{i=1}^{n-1} \text{Ext}^1(\Omega^{i-1}, \Omega^i) \longrightarrow \mathbb{G}_a^{n-1} \xrightarrow{\text{sum}} \mathbb{G}_a \tag{1.3.4}$$

where $\mathcal{E}xt^1(\mathcal{F}, \mathcal{G})$ denotes the extension stack with underlying scheme $\text{Ext}^1(\mathcal{F}, \mathcal{G})$. The first arrow is given by

$$(\mathcal{E}', (\mathcal{E}_i), (\tilde{s}_i), \beta) \mapsto \prod_i (0 \longrightarrow \mathcal{E}_i/\mathcal{E}_{i-1} \longrightarrow \mathcal{E}_{i+1}/\mathcal{E}_{i-1} \longrightarrow \mathcal{E}_{i+1}/\mathcal{E}_i \longrightarrow 0)$$

The third arrow is an isomorphism, deduced from the Serre duality and the last arrow is the sum. For a fixed non-trivial additive character $\psi : \mathbf{k} \rightarrow \mathbb{C}$, let $\mathcal{J}_\psi$ be the Artin-Schreier sheaf on $\mathbb{G}_a$.

The stack Coh_0 of torsion sheaves can be decomposed into connected components $\text{Coh}_0 = \coprod_{d \geq 0} \text{Coh}_0^d$, the Abel-Jacobi map $\text{AJ}^d : X^{(d)} \rightarrow \text{Coh}_0^d$ can be fits into a diagram:

$$\begin{array}{ccccc}
X^d & \longrightarrow & \widetilde{\text{Coh}}_0^d & \xrightarrow{\text{supp}_d} & X^d \\
\downarrow \text{sym} & & \downarrow \pi_d & & \\
X^{(d)} & \xrightarrow{\text{AJ}^d} & \text{Coh}_0^d & &
\end{array} \quad \text{where} \quad \begin{array}{ccccc}
\text{Coh}_0^d(S) & \xleftarrow{\pi_d} & \widetilde{\text{Coh}}_0^d(S) & \xrightarrow{\text{supp}_d} & X^d(S) \\
\downarrow \psi & & \downarrow \psi & & \downarrow \psi \\
\mathcal{T}_d & \longleftarrow & \langle \mathcal{T}_d \twoheadrightarrow \mathcal{T}_{d-1} \twoheadrightarrow \dots \mathcal{T}_0 = 0 \rangle & \longmapsto & \prod_{i=1}^d \text{supp}(\mathcal{T}_i/\mathcal{T}_{i-1})
\end{array}$$

Here the stack $\widetilde{\text{Coh}}_0^d$ can be regarded as iterated extension of torsion sheaves. If we take X to be $\mathbb{A}^1$, then Coh_0^d is isomorphic to the stack $[\mathfrak{gl}_d/\text{GL}_d]$, $\widetilde{\text{Coh}}_0^d$ is isomorphic to $[\widetilde{\mathfrak{gl}}_d/\text{GL}_d]$ and the map π_d above is thus the (simultaneous) Springer resolution. To get a geometric version of W_σ , we first need to translate σ to a local system $\mathbb{L}$ (on X) via monodromy and we define:

Definition 1.10. Let $\mathbb{L}$ be a rank n local system on X , define the **Laumon sheaf** $\mathcal{L}_{\mathbb{L}}$ to be a perverse sheaf on Coh_0 such that its restriction to Coh_0^d is (up to cohomological shift) $\mathcal{L}_{\mathbb{L}}^d := (\pi_{d,*} \text{supp}_d^* \mathbb{L}^{\boxtimes d})^{\text{Sd}}$.

We define the (coherent extended) Whittaker sheaf to be $\widetilde{\mathcal{W}}_{\mathbb{L}} := \alpha^*(\mathcal{L}_{\mathbb{L}}) \otimes \text{ev}^*(\mathcal{J}_\psi)[\dim(\tilde{Q})]$, it can be regarded as sheaf version of W_σ . Recall that the function f'_σ is obtained by push-forward of W_σ along $Q \rightarrow M'_n$, similarly we set $\mathcal{S}'_{\mathbb{L}} := \tilde{\nu}_!(\widetilde{\mathcal{W}}_{\mathbb{L}})$ it is a sheaf on Coh'_n and finally we denote by $\text{Aut}'_{\mathbb{L}}$ the restriction of $\mathcal{S}'_{\mathbb{L}}$ to Bun'_n .

Note that throughout the construction, the stack $\overline{Q}$ is not used. In fact, the actual Whittaker sheaf should be constructed out from there, and is quite crucial for proof of Hecke eigen-property.

Let $\overline{Q}^d$ (resp. Q^d , $\tilde{Q}^d$) denote the connected component of $\overline{Q}$ (resp. Q , $\tilde{Q}$) so that $\mathcal{M}$ in the first coordinate has degree d . Let $\text{ev} : Q \rightarrow \mathbb{G}_a$ denote the composition $\text{ev} \circ \tilde{j}$. We define

$$\mathcal{W}_{\mathbb{L}}^0 := \bar{j}_! \text{ev}^*(\mathcal{I}_{\psi}) [\dim(Q^0)]$$

For a fixed degree $d \geq 0$, the stack of (Hecke) modification is defined by

$$\text{Bun}_n \xleftarrow{h^\leftarrow} \text{Mod}_n^d(S) = \langle (\mathcal{E}, \mathcal{E}', \beta : \mathcal{E} \rightarrow \mathcal{E}') \mid \mathcal{E}'/\mathcal{E} \text{ torsion sheaf of degree } d \rangle \xrightarrow{h^\rightarrow} \text{Bun}_n$$

Let q be the composition $\rho_n^{\text{bun}} \circ \bar{v} : \overline{Q} \rightarrow \text{Bun}_n : (\mathcal{E}, (s_i)) \mapsto \mathcal{E}$. We have the extended Hecke modification diagram via the Zastava space: let $\mathcal{Z}^d := \overline{Q}^0 \times_{\text{Bun}_n}^{\leftarrow} \text{Mod}_n^d$, it maps to $\overline{Q}^d$ as following:

$$'h^\rightarrow : \mathcal{Z}^d \longrightarrow \overline{Q}^d : (\mathcal{E}, (s_i), \mathcal{E}', \beta : \mathcal{E} \rightarrow \mathcal{E}') \mapsto \left(\mathcal{E}', \left(s'_i : \Omega^{\sum_{j=1}^i (n-j)} \xrightarrow{s_i} \Lambda^i \mathcal{E} \xrightarrow{\Lambda^i \beta} \Lambda^i \mathcal{E}' \right) \right)$$

The above map can be viewed as the extended $h^\rightarrow$, more precisely we have the following diagram commutes:

$$\begin{array}{ccccc} \overline{Q}^0 & \xleftarrow{'h^\leftarrow} & \mathcal{Z}^d & \xrightarrow{'h^\rightarrow} & \overline{Q}^0 \\ \downarrow q & & \downarrow 'q & & \downarrow q \\ \text{Bun}_n & \xleftarrow{h^\leftarrow} & \text{Mod}_n^d & \xrightarrow{h^\rightarrow} & \text{Bun}_n \end{array}$$

where $'h^\leftarrow$ and $'q$ are projections. Let $\pi : \text{Mod}_n^d \rightarrow \text{Coh}_0^d, (\mathcal{E}, \mathcal{E}', \beta) \mapsto \mathcal{E}'/\mathcal{E}$.

Definition 1.11. We define the **Whittaker sheaf** to be $\mathcal{W}_{\mathbb{L}} \in D^b(\overline{Q}, \overline{Q}_\ell)$ so that its restriction to $\overline{Q}^d$ is

$$\mathcal{W}_{\mathbb{L}}^d := 'h_! \left('h^{\leftarrow,*}(\mathcal{W}_{\mathbb{L}}^0) \otimes (\pi \circ 'q)^* \left(\mathcal{L}_{\mathbb{L}}^d \right) \right) [dn]$$

In particular when $d = 0$, the Laumon sheaf is trivial, so the notation matches well for $\mathcal{W}_{\mathbb{L}}^0$ over $\overline{Q}^0$.

Theorem 1.12. (i) The restriction of $\mathcal{W}_{\mathbb{L}}^d$ to Q^d is isomorphic to the restriction of $\widetilde{\mathcal{W}}_{\mathbb{L}}^d$.

(ii) The restriction of $S'_{\mathbb{L}}$ is isomorphic to $\bar{v}_!(\mathcal{W}_{\mathbb{L}})$, we use the notation $\text{Aut}'_{\mathbb{L}}$.

(iii) The Whittaker sheaf $\mathcal{W}_{\mathbb{L}}^d$ is a perverse sheaf on $\overline{Q}^d$, moreover, it is isomorphic to $\bar{j}_{!*}(\mathcal{W}_{\mathbb{L}}^d|_{Q^d})$.

(iv) Assume that $\mathbb{L}$ is an irreducible local system corresponds to σ , then the trace of Frobenius of $\mathcal{W}_{\mathbb{L}}$ coincides with the restriction of Whittaker function $W_\sigma|_{\overline{Q}^d(\mathbf{k})}$.

Proposition 1.13. Assume that $\mathbb{L}$ is an irreducible local system corresponds to σ , then the trace of Frobenius of $\text{Aut}'_{\mathbb{L}}$ on $\text{Bun}'_n(\mathbf{k}) = \text{Mir}_n(F) \backslash \dot{P}(\mathbb{A}_F) / \dot{P}(\mathbb{O}_F)$ agrees with the restriction of f'_σ to Bun'_n .

So far we have recovered the Whittaker functions by using geometric construction. To proceed we want to find a way to extend the sheaf to Bun_n . Several more stacks will be introduced.

1.4 Fourier Transform and the construction of automorphic sheaf

Fix any line bundle $\mathcal{L}^{\text{fix}}$, there exists a number $c_{g,n}$ (depends on genus g and rank n) such that for any $\mathcal{M} \in \text{Bun}_n^d$ with degree $d > c_{g,n}$, the condition $\text{Hom}(\mathcal{M}, \mathcal{L}^{\text{fix}}) \neq 0$ implies $\mathcal{M}$ is very unstable. Moreover, if one chooses $\deg(\mathcal{L}^{\text{fix}}) \gg 0$, then $\text{Hom}(\mathcal{M}, \mathcal{L}^{\text{fix}}) = 0, \forall \mathcal{M} \in \text{Bun}_k, k \leq n$ implies $\deg \mathcal{M} > nk(2g-2) + k(k-1)(g-1)$ and $\text{Ext}^1(\Omega^{k-1}, \mathcal{M}) = 0$.

Let C_k^d be the substack of Coh_k^d that $\mathcal{M} \in C_k^d$ satisfies $\text{Hom}(\mathcal{M}, \mathcal{L}^{\text{fix}}) = 0$ and let $C_k = \coprod_{d \geq c_{g,n}} C_k^d$. In particular, when $k = n$, from above we know the complement of $C_n \cap \text{Bun}_n$ in Bun_n is composed by very unstable vector bundles. Define $\mathcal{V}_k$ and its dual $\mathcal{V}_k^\vee$ as the following:

$$\begin{aligned} \mathcal{V}_k(S) &= \langle (\mathcal{M}_k, s_k) \mid \mathcal{M}_k \in C_k(S), s_k \in \text{Hom}(\Omega^{k-1} \boxtimes \mathcal{O}_S, \mathcal{M}_k) \rangle \supseteq \mathcal{V}_k^\circ(S) : s_k \text{ injective}; \\ \mathcal{V}_k^\vee(S) &= \langle (0 \rightarrow \Omega^k \boxtimes \mathcal{O}_S \rightarrow \mathcal{M}_{k+1} \rightarrow \mathcal{M}_k \rightarrow 0) \in \text{Ext}^1(\mathcal{M}_k, \Omega^k) \mid \mathcal{M}_k \in C_k(S) \rangle \supseteq \mathcal{V}_k^{\vee, \circ}(S) : \mathcal{M}_{k+1} \in C_{k+1}(S). \end{aligned}$$

In fact, the duality between $\mathcal{V}_k$ and $\mathcal{V}_k^\vee$ is just from the Serre duality. Also we have the isomorphism $\mathcal{V}_k^\circ \xrightarrow{\cong} \mathcal{V}_{k-1}^\circ$ by sending $(\mathcal{M}_k, s_k)$ to $0 \rightarrow \Omega^{k-1} \xrightarrow{s_k} \mathcal{M}_k \rightarrow \text{Coker}(s_k) \rightarrow 0$. It is easy to verify that $\text{Coker}(s_k) \in C_k$.

Denote by p_k and $p_k^\vee$ the natural projection maps $\mathcal{V}_k \rightarrow C_k$ and $\mathcal{V}_k^\vee \rightarrow C_k$ respectively. They are vector bundle maps (hence smooth) and let $\mathcal{V}_k^d$ and $\mathcal{V}_k^{\vee, d}$ be the preimages of C_k^d , on which the projections are of constant rank.

We have:

$$\begin{array}{ccccccc} & & \mathcal{V}_n & \xleftarrow{j_n} & \mathcal{V}_n^\circ & \cong & \mathcal{V}_{n-1}^{\vee, \circ} \xrightarrow{j_{n-1}^\vee} \mathcal{V}_{n-1}^\vee & & \mathcal{V}_{n-1} & & \dots & & \mathcal{V}_0^\vee & & \\ & \swarrow p_n & & & & & \searrow p_{n-1}^\vee & & \swarrow p_{n-1} & & & & \searrow p_0^\vee & & \\ & C_n & & & & & C_{n-1} & & & & & & & & C_0 = \text{Coh}_0 \end{array}$$

Let $\mathcal{F}_{\mathbb{L}, 1} := (j_0^\vee \circ p_0^\vee)^*(\mathcal{L}_{\mathbb{L}})$ be the sheaf on $\mathcal{V}_1^\circ$ and recursively define

$$\mathcal{F}_{\mathbb{L}, k+1} := \text{Four}_\psi(j_{k,*}(\mathcal{F}_{\mathbb{L}, k}))|_{\mathcal{V}_{k+1}^\circ}$$

where Four_ψ is the Fourier transform (component wise) along the dual vector bundle stacks $\mathcal{V}_{n-1}^\vee \rightarrow C_{n-1} \leftarrow \mathcal{V}_{n-1}$. Note that by construction $\mathcal{V}_n^\circ \subseteq \text{Coh}'_n$, we have the following Lemma:

Lemma 1.14. *The sheaf $\mathcal{F}_{\mathbb{L}, n}$ coincides with the restriction of $\mathcal{S}'_{\mathbb{L}}$ to $\mathcal{V}_n^\circ$.*

Theorem 1.15 (Cleaness property). *(i) For every $k = 1, \dots, n-1$, the canonical map $j_{k,*}\mathcal{F}_{\mathbb{L}, k} \rightarrow j_{k,*}\mathcal{F}_{\mathbb{L}, k}$ is an isomorphism. As a corollary, we find that for any $k = 1, \dots, n$, the restriction of $\mathcal{F}_{\mathbb{L}, i}$ to each connected component of $\mathcal{V}_k^\circ$ is an irreducible perverse sheaf (up to cohomological shift).*

(ii) Let $p_n^\circ := p_n \circ j_n$, then there exist $\mathcal{S}_{\mathbb{L}}^\circ$ over C_n so that the pullback $p_n^{\circ,}(\mathcal{S}_{\mathbb{L}}^\circ) \cong \mathcal{F}_{\mathbb{L}, n}$ up to cohomological shift. Moreover, the restriction of $\mathcal{S}_{\mathbb{L}}^\circ$ to each connected component is a non-zero irreducible perverse sheaf.*

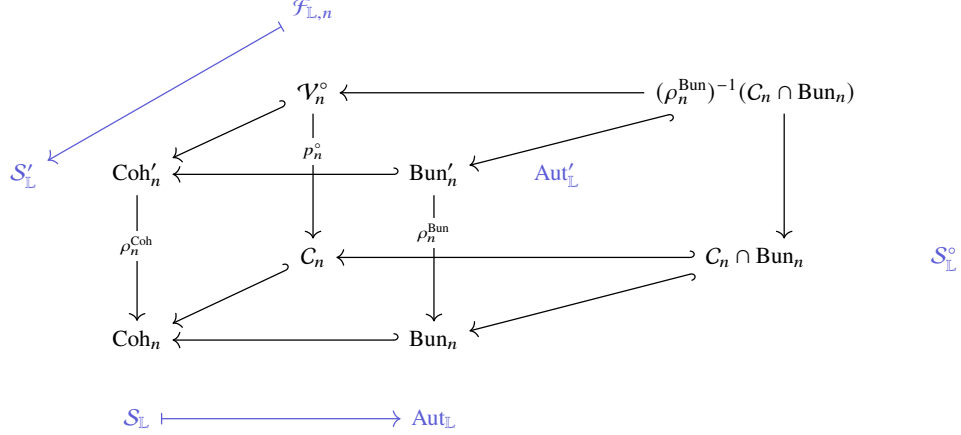
We can see that the perversity of those sheaves $\mathcal{F}_{\mathbb{L}, k}$ essentially comes from the perversity of Laumon sheaf and the general property of Fourier transform. Note that this does NOT guarantee that $\mathcal{S}'_{\mathbb{L}}$ is perverse. In fact those stacks $\mathcal{V}_k$ and C_k are chosen intentionally to make things smooth.

Lemma 1.16. *The restriction of $\mathcal{S}_{\mathbb{L}}^\circ$ to $C_n \cap \text{Bun}_n$ is again perverse. Also, the isomorphism in Lemma 1.14 can be descent to $(\rho_n^{\text{Bun}})^{-1}(C_n \cap \text{Bun}_n)$. Namely $\mathcal{F}_{\mathbb{L}, n}|_{(\rho_n^{\text{Bun}})^{-1}(C_n \cap \text{Bun}_n)} \cong \text{Aut}'_{\mathbb{L}}|_{(\rho_n^{\text{Bun}})^{-1}(C_n \cap \text{Bun}_n)}$.*

We now define:

$$\begin{aligned} \mathcal{S}_{\mathbb{L}} &:= (C_n \hookrightarrow \mathrm{Coh}_n)_{!*} \mathcal{S}_{\mathbb{L}}^{\circ}; \\ \mathrm{Aut}_{\mathbb{L}} &:= \mathcal{S}_{\mathbb{L}}|_{\mathrm{Bun}_n} \end{aligned}$$

What we have constructed so far can be summarized into the following diagram:



Notice that $\mathcal{S}_{\mathbb{L}}$ and $\text{Aut}_{\mathbb{L}}$ are NOT defined entirely on Coh_n (resp. Bun_n) since by construction, C_n only intersects with $\cup_{d \geq c_{g,n}} \text{Coh}_n^d$. So it remains to:

- Find a canonical way to extend $\text{Aut}_{\mathbb{L}}$ to whole Bun_n
- Show that the extended automorphic sheaf $\text{Aut}_{\mathbb{L}}$ is a Hecke-eigensheaf with eigenvalue $\mathbb{L}$ and is cuspidal.

In fact, things are done in [FGV01a] in a reverse order. To elaborate, we first explain the Hecke action in geometric setup: Let Hecke_n^i be the stack so that for $S \in \text{Sch}_{\mathbf{k}}$, it classifies:

$$\text{Hecke}_n^i(S) = \left\langle (x, \mathcal{E}, \mathcal{E}', \beta) \left| \begin{array}{l} x \in X(S), \mathcal{E} \in \text{Bun}_n(S), \mathcal{E}' \in \text{Bun}_n(S); \\ \mathcal{E}'/\mathcal{E} \cong \mathcal{O}_x^{\oplus i}, \beta: \mathcal{E} \hookrightarrow \mathcal{E}' \end{array} \right. \right\rangle$$

with the *Hecke diagram* $\text{Bun}_n \xleftarrow{h_i^-} \text{Hecke}_n^i \xrightarrow{\text{supp} \times h_i^+} X \times \text{Bun}_n$ given by

$$h_i^{\leftarrow} : (x, \mathcal{E}, \mathcal{E}', \beta) \mapsto \mathcal{E}', \quad h_i^{\rightarrow} : (x, \mathcal{E}, \mathcal{E}', \beta) \mapsto \mathcal{E}, \quad \text{supp} : (x, \mathcal{E}, \mathcal{E}', \beta) \mapsto x$$

define the i -th Hecke functor to be $\mathbf{H}_n^i : \mathrm{D}(\mathrm{Bun}_n) \rightarrow \mathrm{D}(X \times \mathrm{Bun}_n) : \mathcal{F} \mapsto (\mathrm{supp} \times h_i^{\rightarrow})_! (h_i^{\leftarrow})^*(\mathcal{F})[(n-i)i]$, this is the geometric analog of global Hecke operator T_n^i . In particular, if we fix $x \in X$ and let $\mathrm{Hecke}_{n,x}^i$ be the fiber of supp over $x \in X$. Then locally over x the Hecke diagram becomes $\mathrm{Bun}_n \xleftarrow{h_{i,x}^{\leftarrow}} \mathrm{Hecke}_{n,x}^i \xrightarrow{h_{i,x}^{\rightarrow}} \mathrm{Bun}_n$, where $h_{i,x}^{\leftarrow} : (\mathcal{E}, \mathcal{E}', \beta) \mapsto \mathcal{E}'$, $h_{i,x}^{\rightarrow} : (\mathcal{E}, \mathcal{E}', \beta) \mapsto \mathcal{E}$ and the local Hecke operator is $\mathbf{H}_{n,x}^i(\mathcal{F}) := \left(h_{i,x}^{\rightarrow}\right)_! (h_{i,x}^{\leftarrow})^*(\mathcal{F})$ which is the geometric analog of T_x^i .

Definition 1.17. Let $\mathbb{L}$ be a rank n irreducible local system over X , for $i = 1, \dots, n$, we say $\mathcal{F} \in \mathbf{D}(\mathbf{Bun}_n)$ satisfies i -th Hecke eigen-property with eigenvalue $\mathbb{L}$ if

$$\mathbf{H}_n^i(\mathcal{F}) = \Lambda^i(\mathbb{L}) \boxtimes \mathcal{F}$$

If for any $i = 1, \dots, n$, $\mathcal{F}$ satisfies the Hecke eigen-property, then we say it is a **Hecke eigensheaf**.

An interesting fact about the Hecke eigen-property is that the i -the eigen-properties follows automatically from the first one.

Proposition 1.18. *If $\mathcal{F}$ satisfies $\mathbf{H}_n^1(\mathcal{F}) = \mathbb{L} \boxtimes \mathcal{F}$ then it is a Hecke eigensheaf.*

Proof. Consider the following diagram:

$$\begin{array}{ccccc}
 \text{Hecke}_n^i & \xrightarrow{\quad h_i^\leftarrow \quad} & \text{Bun}_n & & \\
 \downarrow & \searrow & \nearrow h^\leftarrow & & \uparrow \tilde{h}^\leftarrow \\
 \text{supp} \left(\text{Hecke}_n^{i,+} \right) & \xrightarrow{\quad} & \text{Mod}_n^{-i} & \xleftarrow{\quad \text{fgt} \quad} & \widetilde{\text{Mod}}_n^{-i} \\
 \downarrow & & \downarrow \text{supp} & & \downarrow \widetilde{\text{supp}} \\
 X & \xrightarrow{\quad \Delta \quad} & X^{(i)} & \xleftarrow{\quad \text{Sym} \quad} & X^i
 \end{array} \tag{1.4.1}$$

We have already defined Mod_n^{-i} it has support over the symmetric product $X^{(-i)}$, the stack $\widetilde{\text{Mod}}_n^{-i}$ classifies the iterated inclusions $\langle (\mathcal{E} = \mathcal{E}_0 \subseteq \mathcal{E}_1 \subseteq \dots \subseteq \mathcal{E}_i = \mathcal{E}') \mid \mathcal{E}_k/\mathcal{E}_{k-1} \text{ is a simple skyscraper sheaf} \rangle$. The support map $\widetilde{\text{supp}}$ is the product of support $\mathcal{E}_k/\mathcal{E}_{k-1}$, so that $\text{supp} \circ \text{fgt} = \text{Sym} \circ \widetilde{\text{supp}}$ (forgetting the order of support). Finally $\text{Hecke}_n^{i,+}$ called the positive Hecke stack, is the fiber product in which Hecke_n^i is a closed substack.

Denote $\mathcal{S}pr := \text{fgt}_! \text{IC}_{\widetilde{\text{Mod}}_n^{-i}}[i(n-1)]$, recall $h^\leftarrow, h^\rightarrow : \text{Mod}_n^{-i} \rightarrow \text{Bun}_n$. Also set $\tilde{h}^\leftarrow, \tilde{h}^\rightarrow : \widetilde{\text{Mod}}_n^{-i} \rightarrow \text{Bun}_n$ that maps to the head (the end) of the flag respectively. They are compatible with $h_i^\leftarrow, h_i^\rightarrow$. We have

$$\begin{aligned}
 & (\text{supp} \times h^\rightarrow)_! ((h^\leftarrow)^*(\mathcal{F}) \otimes \mathcal{S}pr) = ((\text{Sym} \circ \widetilde{\text{supp}}) \times \tilde{h}^\rightarrow)_! (\tilde{h}^\leftarrow)^*(\mathcal{F})[i(n-1)] \\
 & = (\text{Sym} \times \text{id})_! \left(\mathbf{H}_n^1 \circ \dots \circ \mathbf{H}_n^1(\mathcal{F}) \right) = \text{Sym}_!(\mathbb{L}^{\boxtimes i}) \boxtimes \mathcal{F}
 \end{aligned}$$

Note that both sides admit S_i -action, consider the isotypic component of sgn-representation, the left becomes $\mathbf{H}_n^i(\mathcal{F})$ while the another side is $\Lambda^i \mathbb{L} \boxtimes \mathcal{F}$. This concludes the proof. $\square$

Back to $\text{Aut}_{\mathbb{L}}$ which is defined on $\cup_{d \geq c_{g,n}} \text{Bun}_n^d$. We have $(\text{supp} \times h^\rightarrow)^{-1}(X \times (C_n \cap \text{Bun}_n)) \subseteq (h^\leftarrow)^{-1}(C_n \cap \text{Bun}_n)$, thus the Hecke operator is well defined on $C_n \cap \text{Bun}_n$. We arrive to the following:

Theorem 1.19. *The restriction of $\text{Aut}_{\mathbb{L}}$ to $C_n \cap \text{Bun}_n$ is a Hecke eigensheaf with eigenvalue $\mathbb{L}$.*

Proof. By Proposition 1.18, we reduce to the case of $\mathbf{H}_n^1$. In fact, we can lift the Hecke diagram to Bun'_n and consider things there:

$$\begin{array}{ccccc}
 \text{Bun}'_n & \xleftarrow{\quad {}''h^\leftarrow \quad} & \text{Bun}'_n \times_{\text{Bun}_n} \text{Hecke}_n^1 & \xrightarrow{\quad \text{supp} \times {}''h^\rightarrow \quad} & X \times \text{Bun}'_n \\
 \downarrow & & \downarrow & & \downarrow \\
 \text{Bun}_n & \xleftarrow{\quad h^\leftarrow \quad} & \text{Hecke}_n^1 & \xrightarrow{\quad \text{supp} \times h^\rightarrow \quad} & X \times \text{Bun}_n
 \end{array}$$

where the arrows above are given by $\text{supp} \times {}''h^\rightarrow : (x, \mathcal{E}, \mathcal{E}', \beta : \mathcal{E}' \hookrightarrow \mathcal{E}, s' : \Omega^{n-1} \rightarrow \mathcal{E}') \mapsto (x, \mathcal{E}', s')$ and ${}''h^\leftarrow : (x, \mathcal{E}, \mathcal{E}', \beta, s') \mapsto (\mathcal{E}, \beta \circ s')$.

Analogous to $\mathbf{H}_n^1$, we define $\mathbf{H}_n^{1'}(\mathcal{F}) := (\text{supp} \times {}''h^{\rightarrow})_! ({}''h^{\leftarrow})^*(\mathcal{F})[n-2]$.

Claim: We have $\mathbf{H}_n^{1'}(\text{Aut}'_{\mathbb{L}}) \cong E \boxtimes \text{Aut}'_{\mathbb{L}}$.

If the Claim holds, then we can restrict ourselves to $C_n \cap \text{Bun}_n$ and $\mathcal{V}_n^{\circ} \cap \rho_n^{-1}(\text{Bun}_n)$, on which ρ_n° is smooth and maps to $\text{Bun}_n \cap C_n$. Apply the smooth base change to the restricted version of above diagram, we get the consequence as described in the Theorem. Note that $\text{Aut}'_{\mathbb{L}} = \bar{v}_!(\mathcal{W}_{\mathbb{L}})$, so prove the Claim, we need to lift again the Hecke diagram. For details, see [FGV01a, Section 7.1 and Appendix A]. $\square$

Theorem 1.20. *The perverse sheaf $\text{Aut}_{\mathbb{L}}$ extends canonically to a Hecke eigensheaf with respect to $\mathbb{L}$ on the entire stack Bun_n .*

Proof. Let us consider the map $\text{mult} : X \times \text{Bun}_n \rightarrow \text{Bun}_n; (x, \mathcal{E}) \mapsto \mathcal{E}(x)$ which increases the degree of vector bundles by n . As a special case ($\mathbf{H}_n^n$) of Proposition 1.18 and using result of Theorem 1.19 we can show that:

$$\text{mult}^*(\text{Aut}_{\mathbb{L}})|_{X \times (C_n \cap \text{Bun}_n)} \cong \Lambda^n \mathbb{L} \boxtimes \text{Aut}_{\mathbb{L}}|_{X \times (C_n \cap \text{Bun}_n)} \quad (1.4.2)$$

And since mult is a smooth map, the above isomorphism extends to entire $\cup_{d \geq c_{g,n}} \text{Bun}_n^d$. To shift between components, we again use the map mult : for any $\text{Bun}_n^{d'}$ and open substack $U \subseteq \text{Bun}_n^{d'}$ of finite type (note $\text{Bun}_n^{d'}$ is locally of finite type), there exists d'' such that for any $x \in X$, $\text{mult}_x^{d''} : \mathcal{E} \mapsto \mathcal{E}(d'' \cdot x)$ sends U into $C_n \cap \text{Bun}_n$. Then we define $\text{Aut}_{\mathbb{L}}|_U$ to be $(\text{mult}_x^{d''})^*(\text{Aut}_{\mathbb{L}}) \otimes (\Lambda^n \mathbb{L}_x)^{\otimes -d''}$. Clearly this does not depend on choice of x , according to 1.4.2, we know the construction does not depend on choice of d'' neither. Finally we extend $\text{Aut}_{\mathbb{L}}$ to entire Bun_n . The Hecke eigen-property follows from a transposition argument of composition of Hecke operators, see [FGV01a, diagram (1.6)]. $\square$

So far we have constructed the automorphic sheaf we want and take the Frobenius trace we get the automorphic form we want. The very natural question comes up is: How can we generalize to reductive groups other than GL_n ? Notice that if we place Bun_n with Bun_G , then the left part of diagram 1.3.3 no longer makes sense, so do several constructions via stacks C_n and $\mathcal{V}_n$ as well as Laumon sheaf. However, it is still plausible to pursue Whittaker sheaves (conceptually because the Casselman-Shalika formula holds for G). That is to say we need to generalize $\bar{Q}$ and Bun'_n in the right part of 1.3.3. The method is so called Drinfeld compactification that we are going to see in the sequel.

2 Whittaker categories

2.1 Drinfeld compactification and Whittaker patterns

We let X be a smooth proper curve and $\mathcal{Y}$ a stack over $\mathbf{k}$, the mapping stack $\mathbf{Maps}(X, \mathcal{Y})$ is a sheaf in groupoids:

$$\mathbf{Maps}(X, \mathcal{Y}) : \mathrm{Sch}_{\mathbf{k}} \longrightarrow \mathrm{Gpd}; \quad S \mapsto \mathcal{Y}(X \times S)$$

For $\mathring{\mathcal{Y}} \subseteq^{\mathrm{open}} \mathcal{Y}$ an open substack, set $\mathbf{Maps}_{\mathrm{gen}}(X, \mathcal{Y} \supseteq \mathring{\mathcal{Y}})$ to be a subsheaf in groupoids of $\mathbf{Maps}(X, \mathcal{Y})$:

$$\mathbf{Maps}_{\mathrm{gen}}(X, \mathcal{Y} \supseteq \mathring{\mathcal{Y}})(S) = \left\{ f \in \mathbf{Maps}(X, \mathcal{Y})(S) \left| \begin{array}{l} \text{for } \forall \bar{s} \rightarrow S \text{ geometric point, there exists} \\ U \subseteq^{\mathrm{open}} X \times \bar{s} \text{ so that } f|_U \text{ factors through } \mathring{\mathcal{Y}} \end{array} \right. \right\}$$

For simplicity, we assume that G has simply-connected derived group. We use the notation $\check{\Lambda}$ for weight lattice $\check{\Lambda}^{\mathrm{pos}}$ for positive weights and $\check{\Lambda}^+$ for dominant ones. Respectively $\Lambda, \Lambda^{\mathrm{pos}}, \Lambda^+$ are co-version.

Definition 2.1. Let Y be a scheme over $\mathbf{k}$, we say that Y is *strongly quasi-affine* if its global section is a finitely generated $\mathbf{k}$ -algebra and there is an open embedding $Y \hookrightarrow \mathrm{Spec}(\Gamma(Y, \mathcal{O}_Y)) =: \bar{Y}$. Call $\bar{Y}$ the *affine closure* of Y .

Lemma 2.2. Let U be any unipotent subgroup of parabolic subgroup P of G , G/U is strongly quasi-affine.

Definition 2.3. Fix B a Borel subgroup of quasi-split reductive group G , and N its unipotent radical, we define the Drinfeld compactification of Bun_B to be:

$$\overline{\mathrm{Bun}}_B := \mathbf{Maps}_{\mathrm{gen}}\left(X, G \backslash \overline{G/N}/T \supseteq \mathbf{BB}\right)$$

Let us compare the difference between Bun_B and $\overline{\mathrm{Bun}}_B$. For $S \in \mathrm{Sch}_{\mathbf{k}}$, $\mathrm{Bun}_B(S)$ classifies the data

$$\left\langle (\mathcal{F}_G, \mathcal{F}_T, \kappa) \left| \begin{array}{l} \mathcal{F}_G \in \mathrm{Bun}_G(S), \mathcal{F}_T \in \mathrm{Bun}_T(S), \kappa = \{\kappa^{\mathcal{V}}\}_{\mathcal{V} \in \mathrm{Rep}(G)} \text{ is a family of subbundles} \\ \text{so that } \kappa^{\mathcal{V}} : (\mathcal{V}^N)_{\mathcal{F}_T} \hookrightarrow \mathcal{V}_{\mathcal{F}_G} \text{ satisfies the Plucker relation.} \end{array} \right. \right\rangle$$

Here we use the Tanakian, in fact a straight forward way to see the data κ would be $\kappa : \mathcal{F}_G = \mathcal{F}_T \overset{T}{\times} G \rightarrow \mathcal{F}_T \overset{M}{\times} G/N$. In comparison, $\overline{\mathrm{Bun}}_B(S)$ classifies the data:

$$\left\langle (\mathcal{F}_G, \mathcal{F}_T, \kappa) \left| \begin{array}{l} \mathcal{F}_G \in \mathrm{Bun}_G(S), \mathcal{F}_T \in \mathrm{Bun}_T(S), \kappa = \{\kappa^{\mathcal{V}}\}_{\mathcal{V} \in \mathrm{Rep}(G)} \text{ family of maps } \kappa^{\mathcal{V}} : (\mathcal{V}^N)_{\mathcal{F}_T} \rightarrow \mathcal{V}_{\mathcal{F}_G} \\ \text{satisfies the Plucker relation and } \forall \bar{s} \rightarrow S \text{ geometric point, restriction of } \kappa^{\mathcal{V}}|_{X_{\bar{s}}} \text{ is injective} \end{array} \right. \right\rangle$$

Equivalently, the data of κ are parametrized by $\check{\Lambda}^+$, we can replace $\mathcal{V}^N$ by $\mathcal{L}_{\mathcal{F}_T}^{\check{\lambda}}$ the line bundle twisted by character $\check{\lambda}$ and $\mathcal{V}$ by $\mathcal{V}_{\mathcal{F}_G}^{\check{\lambda}}$ the vector bundle obtained from the highest weight representation with highest weight $\check{\lambda}$.

Remark 2.4. One can see the difference lies in κ , where we allow the maps to be subsheaf instead of subbundle. This is an approach similar to the one of Laumon's. In fact, when $G = \mathrm{GL}_n$, it is shown by Kuznetsov that Laumon's construction (via Coh stacks) is a small resolution to $\overline{\mathrm{Bun}}_B$.

Note that there are natural projections $\text{Bun}_T = \mathbf{Maps}(X, \mathbf{BT}) \xleftarrow{\mathbf{q}} \overline{\text{Bun}}_B \xrightarrow{\mathbf{p}} \text{Bun}_G = \mathbf{Maps}(X, \mathbf{BG})$. For a fixed T -torsor $\mathcal{F}_T$ on X , we denote by $\overline{\text{Bun}}_N^{\mathcal{F}_T}$ the fiber of $\mathcal{F}_T$ under $\mathbf{q}$. We also call it compactification of Bun_N . Recall the definition in $\overline{Q}$, consider the map

$$\overline{Q}^d \rightarrow X^{(d)} : (\mathcal{E}, (s_i)) \mapsto D := \text{divisor of zeros of } s_n : \Omega^{\frac{n(n-1)}{2}} \rightarrow \Lambda^n \mathcal{E}$$

Let $\overline{Q}^D$ be the preimage of divisor D , then we have the isomorphism: $\overline{Q}^D \cong \overline{\text{Bun}}_N^{\mathcal{F}^D}$, here we consider $G = \text{GL}_n$ and $\mathcal{F}^D$ is the T -torsor chosen to be $(\Omega^{n-1}, \dots, \Omega^1, \mathcal{O}(D))$.

For general group G and any $(\mathcal{F}_G, \kappa) \in \overline{\text{Bun}}_N^{\mathcal{F}_T}$ and each vertex $\mathbf{i}$ in the Dynkin diagram I_G (associated with fundamental weight $\check{\omega}_{\mathbf{i}}$), we can consider the zeros of $\mathcal{L}_{\mathcal{F}_T}^{\check{\omega}_{\mathbf{i}}} \rightarrow \mathcal{V}_{\mathcal{F}_G}^{\check{\omega}_{\mathbf{i}}}$ and their associated divisor on X , denoted by $D_{\mathbf{i}}$. We sum up those divisors by setting $D = \sum_{\mathbf{i} \in I_G} D_{\mathbf{i}} \cdot \alpha_{\mathbf{i}}$. This notation means if we put $\mathcal{F}'_T := \mathcal{F}_T(D)$ then for any $\check{\lambda}$ we have $\mathcal{L}_{\mathcal{F}'_T}^{\check{\lambda}} = \mathcal{L}_{\mathcal{F}_T}^{\check{\lambda}}(\sum_{\mathbf{i}} \langle \alpha_{\mathbf{i}}, \check{\lambda} \rangle \cdot D_{\mathbf{i}})$. This gives us a stratification of $\overline{\text{Bun}}_N^{\mathcal{F}_T}$, labelled by negative coweights ${}_{\gamma} \text{Bun}_N^{\mathcal{F}_T}$, where $\gamma = -\sum_{\mathbf{i} \in I_G} d_{\mathbf{i}} \alpha_{\mathbf{i}} \in -\Lambda^{\text{pos}}$ and ${}_{\gamma} \text{Bun}_N^{\mathcal{F}_T}$ is a vector bundle stack over $\prod_{\mathbf{i} \in I_G} S^{d_{\mathbf{i}}} X$ with fiber at $\underline{x} = (x_{\mathbf{i},1} + \dots + x_{\mathbf{i},d_{\mathbf{i}}})_{\mathbf{i} \in I_G}$ isomorphic to $\text{Bun}_N^{\mathcal{F}'_T}$, where $\mathcal{F}'_T = \mathcal{F}_T(\sum_{\mathbf{i} \in I_G} \sum_{k=1}^{d_{\mathbf{i}}} \alpha_{\mathbf{i}} \cdot x_{\mathbf{i},k})$.

The above construction has a generalization. Namely we can first pin down the data of modification points and weights. Let $\underline{x} = (x_1, \dots, x_n) \in X^n$, $\underline{\nu} = (\nu_1, \dots, \nu_n)$ be n -tuples of points and coweights respectively. Define:

$${}_{\underline{x}, \underline{\nu}} \overline{\text{Bun}}_N^{\mathcal{F}_T}(S) = \left\langle (\mathcal{F}_G, \kappa) \left| \begin{array}{l} \mathcal{F}_G \in \text{Bun}_G(S), \kappa = (\kappa^{\check{\lambda}})_{\check{\lambda} \in \check{\Lambda}^+} \text{ family of embedding of coherent sheaves} \\ \text{where } \kappa^{\check{\lambda}} : \mathcal{L}_{\mathcal{F}_T}^{\check{\lambda}} \hookrightarrow \mathcal{V}_{\mathcal{F}_G}^{\check{\lambda}} \text{ and satisfies Pl\"ucker relations} \end{array} \right. \right\rangle$$

We can easily observe that ${}_{\underline{x}, 0} \overline{\text{Bun}}_N^{\mathcal{F}_T} = \overline{\text{Bun}}_N^{\mathcal{F}_T}$ and that ${}_{\underline{x}, \underline{\nu}} \overline{\text{Bun}}_N^{\mathcal{F}_T} = \overline{\text{Bun}}_N^{\mathcal{F}'_T} = {}_{\underline{x}, 0} \overline{\text{Bun}}_N^{\mathcal{F}'_T}$, where $\mathcal{F}'_T = \mathcal{F}_T(-\sum_k \nu_k \cdot x_k)$, using the notation before. In particular, we can find that if $\underline{\nu}' \geq \underline{\nu}$, i.e. $\nu'_k \geq \nu_k, \forall k = 1, \dots, n$, there is an embedding of stacks ${}_{\underline{x}, \underline{\nu}} \overline{\text{Bun}}_N^{\mathcal{F}_T} \hookrightarrow {}_{\underline{x}, \underline{\nu}'} \overline{\text{Bun}}_N^{\mathcal{F}_T}$, we let ${}_{\underline{x}, \infty} \overline{\text{Bun}}_N^{\mathcal{F}_T} := \varinjlim_{\underline{\nu}} {}_{\underline{x}, \underline{\nu}} \overline{\text{Bun}}_N^{\mathcal{F}_T}$ be the inductive limit. There are two variations, ${}_{\underline{x}, \underline{\nu}} \overline{\text{Bun}}_N^{\mathcal{F}_T}$ by letting $\kappa^{\check{\lambda}}$ to be maximal and ${}_{\underline{x}, \underline{\nu}} \overline{\text{Bun}}_N^{\mathcal{F}_T, \text{gen}}$ so that $\kappa^{\check{\lambda}}$ is regular at neighbor of $\underline{x}$. Thus we have a sequence of embedding:

$${}_{\underline{x}, \underline{\nu}} \overline{\text{Bun}}_N^{\mathcal{F}_T} \xhookrightarrow{\text{open}} {}_{\underline{x}, \underline{\nu}} \overline{\text{Bun}}_N^{\mathcal{F}_T, \text{gen}} \xhookrightarrow{\text{open}} {}_{\underline{x}, \underline{\nu}} \overline{\text{Bun}}_N^{\mathcal{F}_T} \xrightarrow{\text{closed}} {}_{\underline{x}, \infty} \overline{\text{Bun}}_N^{\mathcal{F}_T}$$

The Whittaker category is realized as certain perverse sheaves on the right most stack. The objects are in fact regarded as local ones and endowed with Hecke action, play the role of geometric counterpart of 1.2.1.

We now elaborate the construction. First, there is a generalization of evaluation map 1.3.4 on $\text{Bun}_N^{\mathcal{F}_T}$. Let $\mathbf{i} \in I_G$ be any vertex in the Dynkin diagram, we can associate it with a two dimensional B -representation $V^{\mathbf{i}}$ so that its restriction to $T \subseteq B$ is $\mathbf{1} \oplus \mathbf{1}^{\check{\alpha}_{\mathbf{i}}}$. Here $\mathbf{1}$ is the trivial one dimensional representation and $\mathbf{1}^{\check{\alpha}_{\mathbf{i}}}$ is given by the character $\check{\alpha}_{\mathbf{i}} : T \rightarrow \mathbb{G}_m$ and B acts via the composition $B \rightarrow T \rightarrow \mathbb{G}_m$. Thus for any $(\mathcal{F}_G, \kappa) \in \text{Bun}_N^{\mathcal{F}_T}$ with associated B -torsor on X $\mathcal{F}_B = (\mathcal{F}_G, \mathcal{F}_T, \kappa)$, we have an exact sequence $0 \rightarrow \mathcal{L}_{\mathcal{F}_T}^{\check{\alpha}_{\mathbf{i}}} \hookrightarrow V_{\mathcal{F}_B}^{\mathbf{i}} \rightarrow \mathcal{O}_X \rightarrow 0$, thus a point in extension stack $\mathcal{E}xt(\mathcal{O}_X, \mathcal{L}_{\mathcal{F}_T}^{\check{\alpha}_{\mathbf{i}}})$, which is isomorphic to the quotient stack $\left[H^1(X, \mathcal{L}_{\mathcal{F}_T}^{\check{\alpha}_{\mathbf{i}}}) / H^0(X, \mathcal{L}_{\mathcal{F}_T}^{\check{\alpha}_{\mathbf{i}}}) \right]$.

Let ev_i be the composition $\text{Bun}_N^{\mathcal{F}_T} \rightarrow \mathcal{E}xt(\mathcal{O}_X, \mathcal{L}_{\mathcal{F}_T}^{\check{\alpha}_i}) \rightarrow H^1(X, \mathcal{L}_{\mathcal{F}_T}^{\check{\alpha}_i})$ and let ev be the product:

$$\text{ev} = \prod_{i \in I_G} \text{ev}_i : \text{Bun}_N^{\mathcal{F}_T} \rightarrow \prod_{i \in I_G} H^1(X, \mathcal{L}_{\mathcal{F}_T}^{\check{\alpha}_i})$$

To get a map from $\text{Bun}_N^{\mathcal{F}_T}$ to $\mathbb{G}_a$, we need the data $\varpi = \{\varpi_i : \mathcal{L}_{\mathcal{F}_T}^{\check{\alpha}_i} \hookrightarrow \Omega_X\}$. Set $\text{cond}(\varpi)$ so that $\langle \text{cond}(\varpi), \check{\alpha}_i \rangle =$ divisor of zeroes of ϖ_i called the conductor of ϖ , in other words, it is a function on X with value in $\Lambda_{G_{\text{ad}}}^+$. We have the composition $\text{ev}_{\varpi} : \text{Bun}_N^{\mathcal{F}_T} \xrightarrow{\text{ev}} \prod_{i \in I_G} H^1(X, \mathcal{L}_{\mathcal{F}_T}^{\check{\alpha}_i}) \xrightarrow{\varpi} \prod_{i \in I_G} H^1(X, \Omega_X) \cong \mathbb{G}_a^{|I_G|} \xrightarrow{\text{sum}} \mathbb{G}_a$. We define:

$$\Psi_{\varpi} := (\text{ev}_{\varpi})^*(\mathcal{J}_{\psi})[\dim \text{Bun}_N^{\mathcal{F}_T}], \quad \bar{\Psi}_{\varpi} := \left(J : \text{Bun}_N^{\mathcal{F}_T} \hookrightarrow \overline{\text{Bun}_N^{\mathcal{F}_T}} \right)_{!*} (\Psi_{\varpi})$$

Proposition 2.5. *The intermediate extension $\bar{\Psi}_{\varpi}$ is isomorphic to $\Psi_{\varpi,!} := J_!(\Psi_{\varpi})$ and hence to $\Psi_{\varpi,*} := J_*(\Psi_{\varpi})$.*

In general, we can define similarly those sheaves on ${}_{\underline{x}, \underline{y}}\text{Bun}_N^{\mathcal{F}_T}$ and ${}_{\underline{x}, \underline{y}}\overline{\text{Bun}_N^{\mathcal{F}_T}}$. Let $\text{pr} : \Lambda \rightarrow \Lambda_{G_{\text{ad}}}$ be the projection map of coweight lattices. Fix $\underline{x}$ and $\underline{y}$, we assume that for each k , $\text{cond}_{x_k}(\varpi) + \text{pr}(\nu_k) \in \Lambda_{G_{\text{ad}}}^+$. Under such assumption and the identification ${}_{\underline{x}, \underline{y}}\text{Bun}_N^{\mathcal{F}_T} \cong \text{Bun}_N^{\mathcal{F}_T}$, we can deduce that the induced maps $\varpi'_i : \mathcal{L}_{\mathcal{F}_T}^{\check{\alpha}_i} \rightarrow \Omega$ are regular. Use the construction above, we get $\text{ev}_{\varpi}^{\underline{x}, \underline{y}} := \text{ev}_{\varpi'} : {}_{\underline{x}, \underline{y}}\text{Bun}_N^{\mathcal{F}_T} \rightarrow \mathbb{G}_a$. We define the corresponding sheaves over $\text{Bun}_N^{\mathcal{F}_T}$ and $\overline{\text{Bun}_N^{\mathcal{F}_T}}$ respectively to be:

$$\Psi_{\varpi}^{\underline{x}, \underline{y}} := (\text{ev}_{\varpi}^{\underline{x}, \underline{y}})^*(\mathcal{J}_{\psi})[\dim \text{Bun}_N^{\mathcal{F}_T}], \quad \bar{\Psi}_{\varpi}^{\underline{x}, \underline{y}} := \left(J_{\underline{x}, \underline{y}}^{\infty} : {}_{\underline{x}, \underline{y}}\text{Bun}_N^{\mathcal{F}_T} \hookrightarrow {}_{\underline{x}, \infty}\overline{\text{Bun}_N^{\mathcal{F}_T}} \right)_{!*} (\Psi_{\varpi}^{\underline{x}, \underline{y}})$$

We start with $\mathcal{F}_T$ so that there exists ϖ with conductor $\text{cond}(\varpi) = 0$. Fix one such ϖ , which means fix isomorphisms $\mathcal{L}_{\mathcal{F}_T}^{\check{\alpha}_i} \cong \Omega_X$ (Recall in the constructions 1.3.1 and 1.3.2 we need to fix such Ω -twist). Then we define the **Whittaker category** $\mathcal{W}_{\varpi}^{\underline{x}}$ to be generated by perverse sheaves $\bar{\Psi}_{\varpi}^{\underline{x}, \underline{y}}$ with $\underline{y} \in (\Lambda^+)^n$.

Proposition 2.6. *For any fixed $(\underline{x}, \underline{y})$, under the above assumption on $\text{cond}(\varpi)$, we have the intermediate extension $\bar{\Psi}_{\varpi}^{\underline{x}, \underline{y}}$ being isomorphic to $\Psi_{\varpi,!}^{\underline{x}, \underline{y}} := (J_{\underline{x}, \underline{y}}^{\infty})_!(\Psi_{\varpi}^{\underline{x}, \underline{y}})$ and hence to $\Psi_{\varpi,*}^{\underline{x}, \underline{y}} := (J_{\underline{x}, \underline{y}}^{\infty})_*(\Psi_{\varpi}^{\underline{x}, \underline{y}})$.*

Remark 2.7. In fact, Proposition 2.6 and 2.5 are equivalent. It is clear that Proposition 2.5 implies Proposition 2.6. Conversely, one can first consider the case when $Z(G)$ is connected, in which $\text{pr} : \Lambda \rightarrow \Lambda_{G_{\text{ad}}}$ is surjective. For any ϖ , there exists a divisor D in the form $\sum_{k=1}^n \mu_k \cdot x_k$ such that $\text{pr}(D) = \text{cond}(\varpi)$. We can consider the stack associate with $\mathcal{F}'_T := \mathcal{F}_T(D)$ and data of maps ϖ' with conductor $\text{cond}(\varpi') = 0$. The isomorphism $\text{isom}_{\bullet+\underline{\mu}} : {}_{\underline{x}, \underline{y}}\text{Bun}_N^{\mathcal{F}'_T} \cong {}_{\underline{x}, \underline{y}+\underline{\mu}}\text{Bun}_N^{\mathcal{F}'_T}$ is compatible with evaluation maps, i.e. $\text{ev}_{\varpi} = \text{ev}_{\varpi'} \circ \text{isom}_{\bullet+\underline{\mu}}$. Also we have the closure of ${}_{\underline{x}, \underline{y}}\text{Bun}_N^{\mathcal{F}'_T}$ in ${}_{\underline{x}, \infty}\text{Bun}_N^{\mathcal{F}'_T}$ can be identified with the closure of ${}_{\underline{x}, \underline{y}+\underline{\mu}}\text{Bun}_N^{\mathcal{F}'_T}$ in ${}_{\underline{x}, \infty}\text{Bun}_N^{\mathcal{F}'_T}$. So the result of Proposition 2.6 could be applied to prove Proposition 2.5.

Theorem 2.8. *Under the above assumption on $\text{cond}(\varpi)$, the Whittaker category $\mathcal{W}_{\varpi}^{\underline{x}}$ is semisimple.*

2.2 Geometric Casselman-Shalika

We continue the discussion of the Whittaker category $\mathcal{W}_{\varpi}^{\underline{x}}$. We want to know if this category really gives a geometric version of Whittaker model $\mathbf{W}_{\psi}$. That is to say, we need the Hecke action via Satake category and also

there should be an equivalence between Satake category and Whittaker category, compatible with Hecke action (we can see the semi-simplesness Theorem 2.8 plays an important role here).

To do so, let us fix $\underline{x} = x$ to be just one point. The Hecke diagram locally over x would be:

$$\mathrm{Bun}_G \xleftarrow{h^\leftarrow} \mathrm{Hecke}_{G,x} \xrightarrow{h^\rightarrow} \mathrm{Bun}_G$$

Here $\mathrm{Hecke}_{G,x}(S)$ classifies the data $\left\langle (\mathcal{F}_G, \mathcal{F}'_G, \beta) \mid \mathcal{F}_G, \mathcal{F}'_G \in \mathrm{Bun}_G(S), \beta : \mathcal{F}_G|_{(X-x) \times S} \xrightarrow{\cong} \mathcal{F}'_G|_{(X-x) \times S} \right\rangle$ and the maps send $h^\leftarrow : (\mathcal{F}_G, \mathcal{F}'_G, \beta) \mapsto \mathcal{F}'_G$ and $h^\rightarrow : (\mathcal{F}_G, \mathcal{F}'_G, \beta) \mapsto \mathcal{F}_G$ respectively. By a theorem of Beauville-Laszlo, we know that the affine Grassmannian is the same as Beilinson-Drinfeld grassmannian with modification at x , and is classified by data $(\mathcal{F}_G, \beta)$. Thus the Hecke stack $\mathrm{Hecke}_{G,x}$ can be presented as a twisted product $\mathrm{Hecke}_{G,x} \cong \mathrm{Bun}_G \tilde{\times} \mathrm{Gr}_{G,x}^{\mathrm{BD}}$. We can see there is a stratification on $\mathrm{Hecke}_{G,x}$ which is obtained from the one on $\mathrm{Gr}_{G,x}^{\mathrm{BD}}$. Moreover, the Hecke action over Bun_G is described as the following: for $\mathcal{F}$ over Bun_G and $S \in \mathrm{Sat}$, we define $S \star \mathcal{F} := h_1^\rightarrow(S \tilde{\boxtimes} \mathcal{F})$. Now we can move to ${}_{x,\infty}\overline{\mathrm{Bun}}_N^{\mathcal{F}_T}$, we can lift the Hecke diagram to it:

$$\begin{array}{ccccc} {}_{x,\infty}\overline{\mathrm{Bun}}_N^{\mathcal{F}_T} & \xleftarrow{h^\leftarrow} & Z & \xrightarrow{h^\rightarrow} & {}_{x,\infty}\overline{\mathrm{Bun}}_N^{\mathcal{F}_T} \\ \downarrow & & \downarrow & & \downarrow \\ \mathrm{Bun}_G & \xleftarrow{h^\leftarrow} & \mathrm{Hecke}_{G,x} & \xrightarrow{h^\rightarrow} & \mathrm{Bun}_G \end{array}$$

where $Z := \mathrm{Hecke}_{G,x} \times_{\mathrm{Bun}_G} {}_{x,\infty}\overline{\mathrm{Bun}}_N^{\mathcal{F}_T}$. The left Hecke action by Satake category is given by the following: for $S \in \mathrm{Sat}$, $\mathcal{T} \in \mathrm{Shv}({}_{x,\infty}\overline{\mathrm{Bun}}_N^{\mathcal{F}_T})$, $S \star \mathcal{T} := h_1^\rightarrow(S \tilde{\boxtimes} \mathcal{T})$.

Theorem 2.9 (Geometric Casselman-Shalika). (i) Suppose that $\mathrm{cond}_x(\varpi) = 0$, then we have:

$$\mathbf{IC}_\lambda \star \overline{\Psi}_\varpi^{x,0} \cong \overline{\Psi}_\varpi^{x,\lambda}$$

Here $\mathbf{IC}_\lambda$ is the IC-sheaf on $\overline{\mathrm{Gr}}^\lambda$ and forms simple objects in Sat .

(ii) Combine with Theorem 2.8, we conclude there is a canonical isomorphism between categories

$$\mathrm{Sat} \longrightarrow \mathcal{W}_\varpi^x : S \mapsto S \star \overline{\Psi}_\varpi^{x,0}$$

The category $\mathcal{W}_\varpi^x$ does not depend on the choice of x and ϖ , it is also denoted by $\mathrm{Perv}_N^x(\mathrm{Gr})$ or $\mathrm{Whit}^x(\mathrm{Gr})$. Note that although the objects in $\mathcal{W}_\varpi^x$ are globally defined, they are controlled by local behavior. We need the notion of semi-infinite orbits to give a geometric description of objects in Whittaker category.

Let $\lambda \in \Lambda_G$ be a coweight, we denote by S^λ the orbit of LN on t^λ in affine Grassmannian Gr_G . Note that we have Iwasawa decomposition $LG = \coprod_{\lambda \in \Lambda} LN \cdot t^\lambda \cdot L^+G$, S^λ is thus a stratum in Gr_G , which is infinite dimensional (also of infinite codimension). Moreover, S^λ is locally closed in Gr_G , another orbit $S^\mu \subseteq \overline{S}^\lambda$ if and only if $\lambda - \mu \in \Lambda^{\mathrm{pos}}$. One can see [BR18, Figure 5] that how to visualize the semi-infinite orbits and affine Schubert cells in Bruhat-Tits tree (when $G = \mathrm{SL}_2$). We have the following dimension estimation which plays a crucial role in Mirkovic-Vilonen's proof of geometric Satake equivalence:

Theorem 2.10. [MV07, Theorem 3.2] (i) The intersection $\mathrm{Gr}^\lambda \cap S^\nu$ is empty unless $\nu \in \mathrm{Conv}(W \cdot \lambda) \cap R_G$, where Conv means the convex cone and R_G denotes for the coroot lattice.
(ii) If $\overline{\mathrm{Gr}}^\lambda \cap S^\nu \neq \emptyset$, then it is of pure dimension $\langle \check{\rho}, \lambda + \nu \rangle$.

We are going to introduce the main theorem of [FGV01b] (geometric Casselman-Shalika formula):

Theorem 2.11. [FGV01b, Theorem 1] The stalks of $\mathbf{IC}_\lambda \star \overline{\Psi}^{\mu+\nu}_{\overline{\omega}}$ are computed as the cohomology of the $\frac{\infty}{2}$ -orbit

$$H_c^{\langle \check{\rho}, 2\nu \rangle} \left(\overline{\mathrm{Gr}}^\lambda \cap S^\nu, \mathbf{IC}_\lambda|_{\overline{\mathrm{Gr}}^\lambda \cap S^\nu} \otimes (\chi_\mu^\nu|_{\overline{\mathrm{Gr}}^\lambda \cap S^\nu})^*(\mathcal{I}_\psi) \right)$$

Notice the above convolution product makes sense when $\mu + \nu \in \Lambda^+$, in which case the character $\chi_\mu^\nu : S^\nu \rightarrow \mathbb{G}_a$ is given by $\chi_\mu^\nu : n \cdot t^\nu \mapsto \chi(\mathrm{Ad} t^\mu \cdot n)$ with respect to $\chi : LN \rightarrow \mathbb{G}_a$.

Moreover, the above cohomologies vanish for other degrees (Comparing Theorem 1.2).

3 local/global Whittaker categories, revisited

3.1 Twisted $\mathcal{D}$ -modules and factorizations

The last section basically establish a geometric version for local stories (but uses stack on global automorphic forms) described in section 1.1 and in particular the geometrization of Casselman-Shalika that local Whittaker model is isomorphic to spherical Hecke algebra. The construction in the last section need to fix a point $x \in X$, now we want to let the point x move, so that we can obtain a 'global' version using the feature that the global Whittaker module

$$\mathcal{W}(\psi) := \{W \in C^\infty([G]) \mid W(ng) = \psi(n)W(g)\}$$

can be written as restricted tensor product of local Whittaker models. Also recall that for $\varphi \in C_{\text{cusp}}^\infty([G])$, we can associate it with an element in $\mathcal{W}(\psi)$ by taking

$$\mathcal{W}(\psi) \ni W_\psi^\varphi : g \in [G] \mapsto \int_{[N]} \varphi(ng) \bar{\psi}(n) dn$$

This procedure gives a map $C_{\text{cusp}}^\infty([G]) \rightarrow \mathcal{W}(\psi)$ called **Whittaker coefficient**. The goal of this section has three-folds, we want to give a geometric construction of local Whittaker model purely over a disk, then we want to compare such construction with the one using Drinfeld compactification. Finally, we would like to geometrize the operator of Whittaker coefficient.

To warm up, we introduce how to "glue" local categories over points together, based on the factorization feature. The intuition of the construction certainly comes from the local-global compatibility of Whittaker models. Recall that for any prestack $\mathcal{X}$, we can define its **Ran space** $\text{Ran}(\mathcal{X})$ to be the following colimit:

$$\text{Ran}(\mathcal{X}) := \text{colim}_{I \in \text{fs}} \mathcal{X}^I,$$

where fs means (non-empty) finite sets and the limit is taken over surjective maps. The Ran space defines a sheaf that sends R to set of finite subsets of $\mathcal{X}(R)$ and thus we can view $\text{Ran}(\mathcal{X})$ as a prestack. However, $\text{Ran}(\mathcal{X})$ is far from being (ind-)representable, even if $\mathcal{X}$ is a scheme. Likewise, we can define the Ran version of affine grassmannian to be the following:

$$\text{Gr}_{G, \text{Ran}} : R \mapsto \left\langle (\underline{x}, \mathcal{E}, \beta) \left| \begin{array}{l} \underline{x} \in \text{Ran}(X), \mathcal{E} : G\text{-torsor on } X_R; \\ \beta \text{ trivialization on } X - \underline{x} \end{array} \right. \right\rangle$$

here X is the curve we are considering. Then we have two obvious forgetful maps $\text{Gr}_{G, \text{Ran}} \rightarrow \text{Ran}(X)$ and $\pi_G : \text{Gr}_{G, \text{Ran}} \rightarrow \text{Bun}_G$. The first morphism actually exhibits the factorization natural of $\text{Gr}_{G, \text{Ran}}$: for any finite set I , we can define $\text{Gr}_{G, \mathcal{X}^I}$ to be

$$\text{Gr}_{G, \mathcal{X}^I} : R \mapsto \left\langle (\underline{x}, \mathcal{E}, \beta) \left| \begin{array}{l} \underline{x} \in \mathcal{X}^I, \mathcal{E} : G\text{-torsor on } X_R; \\ \beta \text{ trivialization on } X - \underline{x} \end{array} \right. \right\rangle$$

One can see Gr_{G,X^I} is isomorphic to the fiber product $X^I \times_{\mathrm{Ran}(X)} \mathrm{Gr}_{G,\mathrm{Ran}(X)}$. And we have the following fact about the determinant line bundles, see [Zhu16, Proposition 3.1.17] for more concrete description.

Proposition 3.1. *One can consider the family of determinant line bundles $\mathcal{L}_{\mathrm{det},X^I}$ on Gr_{G,X^I} , then they factorize and give rises to a determinant line bundle $\mathcal{L}_{\mathrm{det},\mathrm{Gr}_G}^{\mathrm{Ran}}$ on $\mathrm{Gr}_{G,\mathrm{Ran}(X)}$.*

Moreover, we can define a determinant line bundle $\mathcal{L}_{\mathrm{det},\mathrm{Bun}_G}$ over Bun_G , normalized so that the stalk of determinant line bundle $\mathcal{L}_{\mathrm{det},\mathrm{Bun}_G}$ on a point $\mathcal{P}_G$, which corresponds to a G -torsor on X , is

$$\det(R\Gamma(X, \mathfrak{g}_{\mathcal{P}_G})) \otimes \det(R\Gamma(X, \mathfrak{g}_{\mathcal{P}_G^0}))^{-1}$$

Notice that stalk of determinant line bundle on Beilinson-Drinfeld Grassmannian has the same expression (the trivialization data does not matter), which tells us that we have the identification

$$\mathcal{L}_{\mathrm{det},\mathrm{Gr}_G}^{\mathrm{Ran}} \cong \pi^*(\mathcal{L}_{\mathrm{det},\mathrm{Bun}_G})$$

From now on, we will only focus on curve over $\mathbb{C}$ and work over the category of $\mathcal{D}$ -modules instead of constructible sheaves. Through the construction of Pfaffian line bundle in [BD, Section 4], we know the choice of the square root $\Omega_X^{\frac{1}{2}}$ of canonical line bundle Ω_X^1 gives a (canonical) square root $\mathcal{L}_{\mathrm{det},\mathrm{Gr}_G}^{\frac{1}{2}}$ of determinant line bundle on Gr_G , as well as $(\mathcal{L}_{\mathrm{det},\mathrm{Gr}_G}^{\mathrm{Ran}})^{\frac{1}{2}}$ on $\mathrm{Gr}_{G,\mathrm{Ran}}$ and $\mathcal{L}_{\mathrm{det},\mathrm{Bun}_G}^{\frac{1}{2}}$ on Bun_G . Then the above construction of square root of determinant line bundles provide us the category of twisted $\mathcal{D}$ -modules. Let $\mathcal{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_G)$ denote the factorization category of $\mathcal{L}_{\mathrm{det},\mathrm{Gr}_G}^{\frac{1}{2}}$ -twisted $\mathcal{D}$ -modules over Gr_G , also $\mathcal{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G,\mathrm{Ran}})$ the category of $(\mathcal{L}_{\mathrm{det},\mathrm{Gr}_G}^{\mathrm{Ran}})^{\frac{1}{2}}$ -twisted $\mathcal{D}$ -modules on $\mathrm{Gr}_{G,\mathrm{Ran}}$. Another way of twisting introduced in [GR14] is de Rham twisting:

Let $\mathcal{Y}$ be a prestack, with associated de Rham prestack $\mathcal{Y}_{\mathrm{dR}} : R \mapsto \mathcal{Y}^{(\mathrm{cl},\mathrm{red})}R$. We have the obvious de Rham projection map $p_{\mathrm{dR}} : \mathcal{Y} \rightarrow \mathcal{Y}_{\mathrm{dR}}$ which is induced by $\mathrm{cl},\mathrm{red}R \rightarrow R$ in $\mathrm{dAff}^{\mathrm{op}}$. Also for any prestack $\mathcal{Y}'$ and any affine group scheme H , we denote by $\mathrm{Gerby}_H(\mathcal{Y}')$ the prestack of H -gerbes Maps $(\mathcal{Y}', B^2H)$. In particular, the de Rham projection induces $p_{\mathrm{dR}}^* : \mathrm{Gerby}_H(\mathcal{Y}_{\mathrm{dR}}) \rightarrow \mathrm{Gerby}_H(\mathcal{Y})$.

Definition 3.2. Define the space of *de Rham twists* on $\mathcal{Y}$ to be

$$\mathrm{Tw}_{\mathcal{Y}}^{\mathrm{dR}} := \mathrm{Ker}(p_{\mathrm{dR}}^* : \mathrm{Gerby}_{\mathbb{G}_m}(\mathcal{Y}_{\mathrm{dR}}) \rightarrow \mathrm{Gerby}_{\mathbb{G}_m}(\mathcal{Y}))$$

More precisely, for $R \in \mathrm{dAff}_{\mathcal{Y}}^{\mathrm{op}}$ with a map $\mathrm{cl},\mathrm{red}R$, $\mathrm{Tw}_{\mathcal{Y}}^{\mathrm{dR}}(R)$ classifies the data of a $\mathbb{G}_m$ -gerbe on $\mathrm{cl},\mathrm{red}R$ and trivialization for each lift $\mathrm{cl},\mathrm{red}R \rightarrow R$. In other words, a $\mathbb{G}_m$ -gerbe on $\mathcal{Y}_{\mathrm{dR}}$ and a trivialization of its pullback to $\mathcal{Y}$. In particular, for any line bundle on $\mathcal{Y}$, we can associate it with a twisting dlog $\mathcal{L}$ by taking trivial $\mathbb{G}_m$ -gerbe on $\mathcal{Y}_{\mathrm{dR}}$ and trivialization to be tensoring with $\mathcal{L}$.

For a prestack $\mathcal{Y}$, we have the ∞ -functor $\mathrm{QCoh}_{\mathcal{Y}} : \mathrm{dAff}_{\mathcal{Y}}^{\mathrm{op}} \rightarrow \mathrm{DGCat}_{\mathrm{cont}}$ which can be right Kan extended to $\mathrm{PreStk}_{\mathcal{Y}}^{\mathrm{op}} \rightarrow \mathrm{DGCat}_{\mathrm{cont}}$. And the category of quasi-coherent sheaves on $\mathcal{Y}$ is exhibited as an (limit of) dg category:

$$\mathrm{QCoh}(\mathcal{Y}) = \lim_{\mathrm{Spec} R \in \mathrm{dAff}_{\mathcal{Y}}^{\mathrm{op}}} \mathrm{Mod}_R$$

Any $\mathbb{G}_m$ -gerbe $\mathcal{Q}$ ($= B\mathbb{G}_m$ -torsor) on $\mathcal{Y}$ acts on $\mathrm{QCoh}_{\mathcal{Y}}$ by tensoring, we denote the twisted functor by $\mathrm{QCoh}_{\mathcal{Y}}^{\mathcal{Q}}$. Thus the data of a twist $T \in \mathrm{Tw}_{\mathcal{Y}}^{\mathrm{dR}}$ give rises to an ∞ -functor $\mathrm{QCoh}_{\mathcal{Y}_{\mathrm{dR}}}^T : \mathrm{dAff}_{/\mathcal{Y}_{\mathrm{dR}}}^{\mathrm{op}} \rightarrow \mathrm{DGCat}_{\mathrm{cont}}$, that restricts to $\mathcal{Y}$ a functor $\mathrm{QCoh}_{\mathcal{Y}}^T : \mathrm{dAff}_{/\mathcal{Y}}^{\mathrm{op}} \rightarrow \mathrm{DGCat}_{\mathrm{cont}}$ and an equivalence: $\mathrm{QCoh}_{\mathcal{Y}}^T \rightarrow \mathrm{QCoh}_{\mathcal{Y}}$. Finally, for a de Rham twisting T on $\mathcal{Y}$, define the T -twisted category of (left) crystals/ $\mathcal{D}$ -modules on $\mathcal{Y}$ to be

$$\mathrm{Crys}^{l,T}(\mathcal{Y}) := \lim_{S \in \mathrm{dAff}_{/\mathcal{Y}_{\mathrm{dR}}}^{\mathrm{op}}} \mathrm{QCoh}^T(S)$$

Remark 3.3. The above construction of de Rham twisting holds for IndCoh . So the twisting is also defined for $\mathcal{D}$ -modules when $\mathcal{Y}$ is not smooth or of finite type.

Proposition 3.4. (i) *The prestack of de Rham twisting is $\mathbf{k}$ -linear. In particular, for any $c \in \mathbf{k}$, we have $c \cdot \mathrm{dlog} \mathcal{L} \in \mathrm{Tw}_{\mathcal{Y}}^{\mathrm{dR}}$. When $c = n \in \mathbb{Z}$, $n \cdot \mathrm{dlog} \mathcal{L} \cong \mathrm{dlog}(\mathcal{L}^{\otimes n})$.*

(ii) *Fix $\mathcal{L}^{\frac{1}{n}}$, a well-defined n -th root of $\mathcal{L}$, consider $\mathcal{D}\text{-mod}_{\frac{1}{n} \cdot \mathrm{dlog} \mathcal{L}}(\mathcal{Y})$ to be the category of de Rham twisted $\mathcal{D}$ -modules on $\mathcal{Y}$, then it is equivalent to $\mathcal{D}\text{-mod}_{\mathcal{L}^{\frac{1}{n}}}(\mathcal{Y})$, the category of $\mathcal{L}^{\frac{1}{n}}$ -twisted $\mathcal{D}$ -modules.*

In particular, if we denote by $\mathcal{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Gr}_G)$ (resp. $\mathcal{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Gr}_{G,\mathrm{Ran}})$) the de Rham twisted $\mathcal{D}$ -modules on Gr_G (resp. $\mathrm{Gr}_{G,\mathrm{Ran}}$) by $\frac{1}{2} \cdot \mathrm{dlog}(\mathcal{L}_{\mathrm{det},\mathrm{Gr}_G})$ (resp. $\frac{1}{2} \cdot \mathrm{dlog}(\mathcal{L}_{\mathrm{det},\mathrm{Gr}_G}^{\mathrm{Ran}})$), we obtain equivalence of categories:

$$\mathcal{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Gr}_G) \xrightarrow{\cong} \mathcal{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_G); \quad \mathcal{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Gr}_{G,\mathrm{Ran}}) \xrightarrow{\cong} \mathcal{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G,\mathrm{Ran}}).$$

Remark 3.5. The construction of μ_2 -gerbe $\mathcal{L}_{\mathrm{det},\mathrm{Gr}_G}^{\frac{1}{2}}$ is NOT compatible with the factorization structure, meaning we no longer have

$$\mathcal{L}_{\mathrm{det},\mathrm{Gr}_{G,X^I \cup X^J}}^{\frac{1}{2}}|_{(X^I \cup X^J)_{\mathrm{disj}}} \cong \mathcal{L}_{\mathrm{det},\mathrm{Gr}_{G,X^I}}^{\frac{1}{2}} \otimes \mathcal{L}_{\mathrm{det},\mathrm{Gr}_{G,X^J}}^{\frac{1}{2}}|_{(X^I \cup X^J)_{\mathrm{disj}}}$$

So one should be careful that even there is an equivalence of categories $\mathcal{D}\text{-mod}_{\mathrm{crit}}(\mathrm{Gr}_{G,\mathrm{Ran}}) \cong \mathcal{D}\text{-mod}(\mathrm{Gr}_{G,\mathrm{Ran}})$ given by trivialization, it is not compatible with factorization structure on both sides.

We also need a twist on affine Grassmannian: $\mathrm{Gr}_{G,\rho(\Omega_X)}$. For $H \rightarrow G$ affine group scheme and $\mathcal{P}_H$ an H -torsor over X , one can formulate a factorization torsor over L^+H by restricting $\mathcal{P}_H$ to a formal disk, denote again this factorization torsor over L^+H by $\mathcal{P}_H$. For any $\mathcal{Y}$ over X with action of L^+H , we can consider the twisted space $\mathcal{Y}_{\mathcal{P}_H} := (\mathcal{Y} \times \mathcal{P}_H) / L^+H$ and the category of $\mathcal{D}$ -modules on it: $\mathcal{D}\text{-mod}(\mathcal{Y})_{\mathcal{P}_H} := \mathcal{D}\text{-mod}(\mathcal{Y}_{\mathcal{P}_H})$. Then $\mathcal{Y}_{\mathcal{P}_H}$ is acted on by twist arc group $(L^+H)_{\mathcal{P}_H}$, with an isomorphism $\mathcal{Y}_{\mathcal{P}_H} / (L^+H)_{\mathcal{P}_H} \cong \mathcal{Y} / L^+H$. We denote by $\left(\mathcal{D}\text{-mod}(\mathcal{Y})^{L^+H} \right)_{\mathcal{P}_H}$ the category of $\mathcal{D}$ -modules over $\mathcal{Y}_{\mathcal{P}_H} / (L^+H)_{\mathcal{P}_H}$. Because of the above isomorphism, we have an equivalence of categories: $\mathcal{D}\text{-mod}(\mathcal{Y})^{L^+H} \cong \left(\mathcal{D}\text{-mod}(\mathcal{Y})^{L^+H} \right)_{\mathcal{P}_H}$.

We can take $H = T$ and $\mathcal{P}_H$ to be the T -torsor $\rho(\Omega_X) := 2\rho(\Omega_X^{\frac{1}{2}})$, which is determined by the coweight $2\rho : \mathbb{G}_m \rightarrow T$ and choice of $\Omega_X^{\frac{1}{2}}$. The prestack acted on by L^+T is taken to be Gr_G and the result prestack is denoted by $\mathrm{Gr}_{G,\rho(\Omega_X)}$. Let $\mathcal{P}_{G,\Omega_X}$ denote the G -torsor over X induced from $\rho(\Omega_X)$, then $\mathrm{Gr}_{G,\rho(\Omega_X)}$ is the functor

$$\mathrm{Sch}_{\mathbf{k}} \ni S \mapsto \left\langle (x, \mathcal{P}_G, \beta) \left| \begin{array}{l} x \in X(S), \mathcal{P}_G : G\text{-torsor over } X_S \\ \beta : \mathcal{P}_G|_{X_S - \{x\}} \xrightarrow{\cong} \mathcal{P}_{G,\Omega_X}|_{X_S - \{x\}} \end{array} \right. \right\rangle$$

Note that one can also define a Ran version twisted affine Grassmannian. Combine with the above definition of twisted $\mathcal{D}$ -modules, we can now make sense of the following category:

$$\mathbf{C} := \mathcal{D}\text{-mod}_{\frac{1}{2}}(\mathrm{Gr}_{G,\rho(\Omega_X)})$$

Use the notation in section 1.3, we can view $\mathbf{C}$ as categorification of functions on $\dot{G}(F_X)/\dot{G}(O_X)$. Similar to $\mathrm{Gr}_{G,\rho(\Omega_X)}$, we can also define $LG_{\rho(\Omega_X)}$ and $LN_{\rho(\Omega_X)} \subseteq LG_{\rho(\Omega_X)}$, where L^+T acts trivially on LN , both $LG_{\rho(\Omega_X)}$ and $LN_{\rho(\Omega_X)}$ act on $\mathrm{Gr}_{G,\rho(\Omega_X)}$. Given a generic character ψ_0 on $N/[N, N]$, similar to the construction in section 2.1, there is a canonical character $\psi : LN_{\rho(\Omega_X)} \rightarrow \mathbb{G}_a$, given by the composition:

$$\psi : LN_{\rho(\Omega_X)} \rightarrow L(N/[N, N])_{\rho(\Omega_X)} \cong \prod_{i \in I_G} (LN_{\alpha_i})_{\rho(\Omega_X)} \xrightarrow{\mathrm{res}} \prod_{i \in I_G} N_{\alpha_i} \xrightarrow{\lambda_0} \mathbb{G}_a$$

In the setting over $\mathrm{char}(\mathbf{k}) = 0$, we no longer have Artin-Schreier sheaves, the substitute is the exponential $\mathcal{D}$ -module $\exp \in \mathcal{D}\text{-mod}(\mathbb{G}_a)$, defined as $\mathcal{D}_{\mathbb{A}^1}/(\partial_z - \mathrm{id})\mathcal{D}_{\mathbb{A}^1}$.

3.2 local/global Whittaker category

Now we want to replace the $\dot{N}(F_X)$ -invariant functions to be LN -(co)invariant $\mathcal{D}$ -modules (with respect to $\psi^*(\exp)$). It remains two question to tackle. First, LN is not an ind-affine group scheme of ind-finite type, so the naive definition of invariant category in the form of $\mathcal{D}\text{-mod}(\mathcal{Y}/H)$ does not work, we need to develop the notion of $\mathcal{D}$ -modules for LN . Then, we need to define $(LN, \psi^*(\exp))$ -invariant category. We refer to [Ber17] for both constructions.

Notice that for H affine group scheme, its loop group LH is an ind-pro-scheme, here by pro-scheme we mean a filtered limit of scheme of finite type, with the limit presented by smooth affine surjective maps. Let $Z = \lim_i Z_i$ be a pro-scheme with arrows $\pi_{i>j} : Z_i \rightarrow Z_j$. Define the category of crystals or $*$ - $\mathcal{D}$ -modules over Z to be

$$\mathcal{D}\text{-mod}_*(Z) := \lim_{i, \pi_*} \mathcal{D}\text{-mod}(Z_i) \cong \mathrm{colim}_{i, \pi^*} \mathcal{D}\text{-mod}(Z_i)$$

The last equality is due to the smoothness of $\pi_{i,j}$, we have $(\pi_{i,j})_*$ being right adjoint of $\pi_{i,j}^*$. We can view $\mathcal{D}\text{-mod}_*$ as a functor of crystals on the category of pro-schemes $\mathrm{Sch}^{\mathrm{pro}}$, which is a left Kan extension of $\mathcal{D}\text{-mod} : \mathrm{Sch}^{\mathrm{ft}} \rightarrow \mathrm{DGCat}$, along the inclusion $\mathrm{Sch}^{\mathrm{ft}} \hookrightarrow \mathrm{Sch}^{\mathrm{pro}}$. Moreover, $\mathcal{D}\text{-mod}_*(Z)$ is dualizable, with dual category $\mathcal{D}\text{-mod}^!(Z) := \mathrm{colim}_{i, \pi^!} \mathcal{D}\text{-mod}(Z_i)$.

Now for ind-schemes of ind-finite type that are exhibited as colimit of closed embeddings, we can also define the functor of crystals as left Kan extension of functor of crystals $\mathcal{D}\text{-mod} : \mathrm{Sch}^{\mathrm{ft}} \rightarrow \mathrm{DGCat}$ as the following:

$$\mathcal{D}\text{-mod} : \mathrm{Ind}^{cl} \mathrm{Sch}^{\mathrm{ft}} \rightarrow \mathrm{DGCat}$$

Furthermore, let $\mathrm{Ind}^{cl} \mathrm{Sch}^{\mathrm{pro}}$ be the category of ind-pro-schemes, i.e. the filtered colimits of pro-schemes under

finitely presented closed embeddings. We can define two functors of crystals:

$$\begin{aligned}\mathcal{D}\text{-mod}_* &: \text{Ind}^{cl} \text{Sch}^{\text{pro}} \rightarrow \text{DGCat} \quad \text{left Kan extension} \\ \mathcal{D}\text{-mod}^! &: \text{Ind}^{cl} \text{Sch}^{\text{pro}} \rightarrow \text{DGCat} \quad \text{right Kan extension}\end{aligned}$$

Now we have both $\mathcal{D}\text{-mod}_*$ and $\mathcal{D}\text{-mod}^!$ well-defined for ind-pro-schemes, in particular, we can apply to loop groups: for H affine group scheme, consider the quotient map $q : LH \rightarrow \text{Gr}_H$. We know that Gr_H can be written as ind-scheme $\text{Gr}_H = \text{colim}_i Z_i$. And thus $LH = \text{colim}_i q^{-1}(Z_i)$. Each $q^{-1}(Z_i)$ can be further written as pro-scheme: for each $j \in \mathbb{N}$, consider the group scheme $(L^+H)_j$ that represents $G(O/t^j)$, then we have

$$q^{-1}(Z_i) = \lim_j Z_i^j := \lim_j Z_i \times_{\text{Gr}_H} (LH/(L^+H)_j)$$

It is clear to see for $i > j$, $LH/(L^+H)_i H \rightarrow LH/(L^+H)_j$ is smooth affine projection, thus $LH = \text{colim}_i \lim_j Z_i^j \in \text{Ind}^{cl} \text{Sch}^{\text{pro}}$.

Proposition 3.6. *Let H be an affine group scheme and LH its loop group, then there is an equivalence $\mathcal{D}\text{-mod}_*(LH) \otimes \mathcal{D}\text{-mod}_*(LH) \cong \mathcal{D}\text{-mod}_*(LH \times LH)$ and a monoidal structure induced by multiplication $(m_{LH})_* : \mathcal{D}\text{-mod}_*(LH \times LH) \rightarrow \mathcal{D}\text{-mod}_*(LH)$. Similarly, the comonoidal structure on $\mathcal{D}\text{-mod}^!(LH)$ is given by $(m_{LH})^!$*

According to Proposition 3.6, we can say a category C is acted on by ind-pro-group scheme $\mathcal{G}$ if there is an action of the monoidal category $(\mathcal{D}\text{-mod}_*(\mathcal{G}), m_*)$. Similarly, we can define the coaction of $(\mathcal{D}\text{-mod}^!(\mathcal{G}), m^!)$. We use the notation $\mathcal{D}_*(\mathcal{G})\text{-mod}$ for the ∞ -category of DG categories that acted on by $\mathcal{D}\text{-mod}_*(\mathcal{G})$. The trivial categorical action is given by $\mathcal{D}\text{-mod}_*(\mathcal{G}) \rightarrow \mathbf{Vect}$. Recall that if V is a representation of group G , then G -invariant part V^G is given by $\text{Hom}_G(\text{triv}, V)$ and G -coinvariant part V_G is given by $\text{triv} \otimes_G V$. Using the same idea, assume C is acted on by $\mathcal{D}\text{-mod}_*(\mathcal{G})$, we define the invariant/coinvariant categories to be:

$$C^{\mathcal{G}} := \text{Hom}_{\mathcal{D}_*(\mathcal{G})\text{-mod}}(\mathbf{Vect}, C); \quad C_{\mathcal{G}} := \mathbf{Vect} \otimes_{\mathcal{D}_*(\mathcal{G})\text{-mod}} C$$

The trivial action of $\mathcal{D}\text{-mod}_*(\mathcal{G})$ on $\mathbf{Vect}$ can be twisted: fix $\mathcal{F} \in \mathcal{D}\text{-mod}^!(\mathcal{G}) = (\mathcal{D}\text{-mod}_*(\mathcal{G}))^\vee$, we can associate it with a functor $\star_{\mathcal{F}} : \mathcal{D}\text{-mod}_*(\mathcal{G}) \otimes \mathbf{Vect} \rightarrow \mathbf{Vect} : (\mathcal{E}, V) \mapsto \langle \mathcal{E}, \mathcal{F} \rangle V$, where $\langle -, - \rangle$ is the pairing between dual categories. To make the functor $\star_{\mathcal{F}}$ an action of $\mathcal{D}\text{-mod}_*(\mathcal{G})$ on C , we need $\mathcal{E}_1 \star_{\mathcal{F}} (\mathcal{E}_2 \star_{\mathcal{F}} V) = (\mathcal{E}_1 * \mathcal{E}_2) \star_{\mathcal{F}} V$. For this, we need $\langle m_*(\mathcal{E}_1 \boxtimes \mathcal{E}_2), \mathcal{F} \rangle = \langle \mathcal{E}_1, \mathcal{F} \rangle \otimes \langle \mathcal{E}_2, \mathcal{F} \rangle$, which is satisfied if

$$m^!(\mathcal{F}) = \mathcal{F} \boxtimes \mathcal{F}; \quad \text{unit}^!(\mathcal{F}) = \mathbf{k}$$

Such $\mathcal{F}$ is called *character $\mathcal{D}$ -module*. In particular, we know [Ber17, Lemma 2.4.3]

Lemma 3.7. *Let $\chi : \mathcal{G} \rightarrow \mathbb{G}_a$ be an additive character, then $\chi^!(\text{exp})$ is a character $\mathcal{D}$ -module.*

We let $\mathbf{Vect}_{\chi}$ denote the category $\mathbf{Vect}$ with $\chi^!(\text{exp})$ -twisted action. For C acted on by $\mathcal{G}$, we define $(\mathcal{G}, \chi)$ -

invariant and $(\mathcal{G}, \chi)$ -coinvariant categories to be

$$C^{\mathcal{G}, \chi} := \text{Hom}_{\mathcal{D}_*(\mathcal{G})-\text{mod}}(\mathbf{Vect}_\chi, C); \quad C_{\mathcal{G}, \chi} := \mathbf{Vect}_\chi \otimes_{\mathcal{D}_*(\mathcal{G})-\text{mod}} C$$

Theorem 3.8. [[Ras21a](#), Theorem 2.1.1] *Let C be acted on by LN , with ψ generic character of LN , then there is an equivalence*

$$C^{LN, \psi} \xrightarrow{\cong} C_{LN, \psi}$$

Definition 3.9. We define the **!-local Whittaker category** to be the ψ -twisted $LN_{\rho(\Omega_X)}$ -invariant category

$$\text{Whit}^!(G) := \mathbf{C}^{LN_{\rho(\Omega_X)}, \psi} = \mathcal{D}\text{-mod}_{\frac{1}{2}}(\text{Gr}_{G, \rho(\Omega_X)})^{LN_{\rho(\Omega_X)}, \psi}$$

define the **!-local Whittaker category** to be the ψ -twisted $LN_{\rho(\Omega_X)}$ -coinvariant category

$$\text{Whit}_*(G) := \mathbf{C}_{LN_{\rho(\Omega_X)}, \psi} = \mathcal{D}\text{-mod}_{\frac{1}{2}}(\text{Gr}_{G, \rho(\Omega_X)})_{LN_{\rho(\Omega_X)}, \psi}$$

We can deduce as a Corollary of Theorem 3.8 that we have an equivalence $\text{Whit}^!(G) \cong \text{Whit}_*(G)$.

The local Whittaker category has factorization nature: for any finite set I , we can consider Gr_{G, X^I} and its twisted version $\text{Gr}_{G, \rho(\Omega_X), X^I}$, then the above construction can be generalized to define $\text{Whit}^!(G)_{X^I}$, respectively $\text{Whit}_*(G)_{X^I}$. The geometric Casselman-Shalika formula in Theorem 2.9 now becomes:

Theorem 3.10. *There is an equivalence of factorization categories:*

$$\text{Whit}^!(G) \xrightarrow{\cong} \text{Rep}(\check{G}); \quad \text{Whit}^!(G)_{\text{Ran}} \xrightarrow{\cong} \text{Rep}(\check{G})_{\text{Ran}}$$

3.3 global Whittaker category

To motivate, recall our construction of Whittaker category in section 2 involves the stack $\overline{\text{Bun}}_N$ that encodes information about global automorphic forms (via the projection to Bun_G). One observation is the following: fix a point $x \in X$, the space of global Whittaker function is $\text{Funct}((N(\mathbb{A}), \psi) \backslash G(\mathbb{A})/G(\mathbb{O}))$, we consider its subspace of functions supported on $G(F_x) \times N(\mathbb{A}^x)G(\mathbb{O}^x)$, where $\mathbb{A}^x$ is set to be $\mathbb{A}/F_x$ (similar for $\mathbb{O}^x$). This subspace equals to

$$((N(\mathbb{A}), \psi) \backslash G(\mathbb{A})/G(\mathbb{O}))$$

which is the space of local Whittaker functions. Moreover, we can consider more generally that allowing level structure (=ramification) at x . More precisely, let K_n be the congruence group of level n at x , i.e kernel of $G(\mathcal{O}_x) = L^+G \twoheadrightarrow G(\mathcal{O}_x/t^n) = (L^+G)_n$, we can replace above to be

$$\text{Funct}((N(\mathbb{A}), \psi) \backslash G(\mathbb{A})/K_n \times G(\mathbb{O}^x))_0 \subseteq \text{Funct}((N(\mathbb{A}), \psi) \backslash G(\mathbb{A})/K_n)$$

where the subscript on the LHS means supported on $G(F_x) \times N(\mathbb{A}^x)G(\mathbb{O}^x)$. The LHS is also a subspace of

$$\text{Funct}(N(F) \backslash G(F_x) \times N(\mathbb{A}^x)/K_n \times N(\mathbb{O}^x))$$

determined by Whittaker condition. The geometrization of above function space would be

$$\mathcal{D}\text{-mod} \left({}_{x,\infty}\overline{\text{Bun}}_N^{G\text{-level}_{n,x}} \right)$$

where in the bracket the prestack is a generalization of ${}_{x,\infty}\overline{\text{Bun}}_N$ in section 2.1 that allows level structure, its definition is quite straightforward, we can consider the stack $\text{Bun}_G^{G\text{-level}_{n,x}}$ of G -bundles with n -level structure at x , which has a projection map $\text{Bun}_G^{G\text{-level}_{n,x}} \rightarrow \text{Bun}_G$ as $(L^+G)_n$ -torsor. Then we define:

$${}_{x,\infty}\overline{\text{Bun}}_N^{G\text{-level}_{n,x}} := {}_{x,\infty}\overline{\text{Bun}}_N \times_{\text{Bun}_G} \text{Bun}_G^{G\text{-level}_{n,x}}$$

Recall in section 2.1 we defined ${}_{x,\infty}\overline{\text{Bun}}_N^{\mathcal{F}_T}$ for a fixed T -torsor $\mathcal{F}_T$, here in the above fiber product ${}_{x,\infty}\overline{\text{Bun}}_N$ we take the T -torsor to be $\rho(\Omega_X) := (\Omega_X^{\frac{1}{2}})^{\otimes m, 2\rho} \times T$ as before. Namely ${}_{x,\infty}\overline{\text{Bun}}_N$ classifies the data of

$$\left((\mathcal{F}_G, \kappa) \left| \begin{array}{l} \mathcal{F}_G \in \text{Bun}_G, \kappa = (\kappa^{\check{\lambda}})_{\check{\lambda} \in \check{\Lambda}^+} \text{ family of embedding of coherent sheaves} \\ \text{where } \kappa^{\check{\lambda}} : (\Omega_X^{\frac{1}{2}})^{\langle \check{\lambda}, 2\rho \rangle} \hookrightarrow \mathcal{V}_{\mathcal{F}_G}^{\check{\lambda}}(\infty \cdot x) \text{ and satisfies Pl\"ucker relations} \end{array} \right. \right)$$

Now we want an action of (LN, ψ) . There is an obstruction to do so: points in ${}_{x,\mu}\overline{\text{Bun}}_N \subseteq {}_{x,\infty}\overline{\text{Bun}}_N$ could have defect supported on $(X - x)^\lambda$, meaning that $\kappa^\lambda : (\Omega_X^{\frac{1}{2}})^{\langle \check{\lambda}, 2\rho \rangle} \hookrightarrow \mathcal{V}_{\mathcal{F}_G}^{\check{\lambda}}(\langle \mu, \check{\lambda} \cdot x \rangle)$ factors through $(\Omega_X^{\frac{1}{2}})^{\langle \check{\lambda}, 2\rho \rangle}(D)$ where $D \in \text{Div}_{(X-x)}^{\deg=\langle \lambda, \check{\lambda} \rangle}$. We use $({}_{x,\mu}\overline{\text{Bun}}_N)_{\text{def}=\lambda}$ to denote the such substack with defect λ . Next we need first to justify the "region" of defect-free by adding auxiliary points: let $\underline{y} = (y_1, \dots, y_k)$ be a collection of points in $X - x$, we use the notation

$$({}_{x,\infty}\overline{\text{Bun}}_N)_{\text{good at } \underline{y}}$$

to denote the substack of ${}_{x,\infty}\overline{\text{Bun}}_N$ that $\kappa^{\check{\lambda}}$ are subbundle maps over the neighbor of y_i 's. A point in $({}_{x,\infty}\overline{\text{Bun}}_N)_{\text{good at } \underline{y}}$ gives B -torsor structure at neighbor of $\underline{y}$, so that the induced T -torsor coincides with $\rho(\Omega_X)$. Let $\mathcal{F}_{B,\Omega_X}^\rho$ denote the induced B -torsor $\rho(\Omega_X)^T \times B$ and define a prestack $({}_{x,\infty}\overline{\text{Bun}}_N)_{\text{good at } \underline{y}}^{N\text{-level}_{\infty,\underline{y}}}$ that classifies data in $({}_{x,\infty}\overline{\text{Bun}}_N)_{\text{good at } \underline{y}}$ with the data of isomorphisms between B -torosr on $\underline{y}$ with $\mathcal{F}_{B,\Omega_X}^\rho$. We obtain an obvious projection map

$$({}_{x,\infty}\overline{\text{Bun}}_N)_{\text{good at } \underline{y}}^{N\text{-level}_{\infty,\underline{y}}} \rightarrow ({}_{x,\infty}\overline{\text{Bun}}_N)_{\text{good at } \underline{y}}$$

which is a $(L^+N)_{\underline{y}}$ -torsor. So we have the following equivalence of categories

$$\mathcal{D}\text{-mod} \left(({}_{x,\infty}\overline{\text{Bun}}_N)_{\text{good at } \underline{y}} \right) \cong \mathcal{D}\text{-mod} \left(({}_{x,\infty}\overline{\text{Bun}}_N)_{\text{good at } \underline{y}}^{N\text{-level}_{\infty,\underline{y}}} \right)^{(L^+N)_{\underline{y}}}$$

We can regard $\mathcal{D}\text{-mod} \left(({}_{x,\infty}\overline{\text{Bun}}_N)_{\text{good at } \underline{y}} \right)$ as geometrization of a subspace of

$$\text{Funct} \left(N(F) \backslash G(F_x) \times N(\mathbb{A}^x) / N(\mathbb{O}^y) \right)$$

The natural action of $(L^+N)_{\underline{y}}$ on $\left(x, \infty \overline{\text{Bun}}_N\right)_{\text{good at } \underline{y}}^{N\text{-level}_{\infty, \underline{y}}}$ can be extended to an action of $(LN)_{\underline{y}}$.

Lemma 3.11. *Consider the strata of $\left(x, \infty \overline{\text{Bun}}_N\right)_{\text{good at } \underline{y}}^{N\text{-level}_{\infty, \underline{y}}}$ that is given by $\left(x, \mu \overline{\text{Bun}}_N\right)_{\text{def}=\lambda, \text{good at } \underline{y}}^{N\text{-level}_{\infty, \underline{y}}}$ and the composition of forgetful maps:*

$$\left(x, \mu \overline{\text{Bun}}_N\right)_{\text{def}=\lambda, \text{good at } \underline{y}}^{N\text{-level}_{\infty, \underline{y}}} \rightarrow \left(x, \mu \overline{\text{Bun}}_N\right)_{\text{def}=\lambda, \text{good at } \underline{y}} \rightarrow \left(X - \{x \cup \underline{y}\}\right)^\lambda$$

Then $(LN)_{\underline{y}}$ acts transitively on the fibers of the above map.

Remark 3.12. We can add level structure at x on $\left(x, \infty \overline{\text{Bun}}_N\right)_{\text{good at } \underline{y}}$ for free, by taking the fiber product

$$\left(x, \infty \overline{\text{Bun}}_N\right)_{\text{good at } \underline{y}}^{G\text{-level}_{n, x}} := \left(x, \infty \overline{\text{Bun}}_N\right)_{\text{good at } \underline{y}} \times_{\text{Bun}_G} \text{Bun}_G^{G\text{-level}_{n, x}}$$

Similar for definitions of

$$\left(x, \infty \overline{\text{Bun}}_N\right)_{\text{good at } \underline{y}}^{N\text{-level}_{\infty, \underline{y}}; G\text{-level}_{n, x}}; \quad \left(x, \mu \overline{\text{Bun}}_N\right)_{\text{def}=\lambda, \text{good at } \underline{y}}^{G\text{-level}_{n, x}}; \quad \left(x, \mu \overline{\text{Bun}}_N\right)_{\text{def}=\lambda, \text{good at } \underline{y}}^{N\text{-level}_{\infty, \underline{y}}; G\text{-level}_{n, x}}$$

And the map in Lemma 3.11 becomes

$$\left(x, \mu \overline{\text{Bun}}_N\right)_{\text{def}=\lambda, \text{good at } \underline{y}}^{N\text{-level}_{\infty, \underline{y}}} \rightarrow \left(x, \mu \overline{\text{Bun}}_N\right)_{\text{def}=\lambda, \text{good at } \underline{y}} \rightarrow \left((L^+G)_{n, x} \times^{(L^+T)_x} \rho(\Omega_X^{\frac{1}{2}})(\mu \cdot x)\right) \times \left(X - \{x \cup \underline{y}\}\right)^\lambda$$

Definition 3.13. Fix auxiliary points $\underline{y}$, define the global Whittaker category at $\underline{y}$ to be the full subcategory of $\mathcal{D}\text{-mod}\left(\left(x, \infty \overline{\text{Bun}}_N\right)_{\text{good at } \underline{y}}\right)$ such that corresponds to

$$\mathcal{D}\text{-mod}\left(\left(x, \infty \overline{\text{Bun}}_N\right)_{\text{good at } \underline{y}}^{N\text{-level}_{\infty, \underline{y}}}\right)^{(LN)_{\underline{y}}, \psi_{\underline{y}}}$$

under the isomorphism $\mathcal{D}\text{-mod}\left(\left(x, \infty \overline{\text{Bun}}_N\right)_{\text{good at } \underline{y}}\right) \cong \mathcal{D}\text{-mod}\left(\left(x, \infty \overline{\text{Bun}}_N\right)_{\text{good at } \underline{y}}^{N\text{-level}_{\infty, \underline{y}}}\right)^{(L^+N)_{\underline{y}}}$. We will denote the corresponding category by $\text{Whit}\left(\left(x, \infty \overline{\text{Bun}}_N\right)_{\text{good at } \underline{y}}\right)$.

Remark 3.14. The same machinery works for stacks like $\left(x, \mu \overline{\text{Bun}}_N\right)_{\text{good at } \underline{y}}$, $\left(x, \leq \mu \overline{\text{Bun}}_N\right)_{\text{good at } \underline{y}}$ to define It also works for the version with G -level structure at x .

Note that in general if C is a category acted on by ind-pro-group scheme $\mathcal{G}$, then we have the forgetful functor $\mathbf{oblv}^{\mathcal{G}, \chi} : C^{\mathcal{G}, \chi} \rightarrow C$. In fact, it admits a right adjoint (averaging functor) $\mathbf{Av}_*^{\mathcal{G}, \chi} : C \rightarrow C^{\mathcal{G}, \chi}$, for simplicity, we assume C is sheaf/ $\mathcal{D}$ -module category $\mathcal{D}\text{-mod}(Y)$ of some ind-scheme $\mathcal{Y}$, such that we have presentation

$$\mathcal{Y} := \text{colim}_i Y_i; \quad \mathcal{G} = \text{colim}_\alpha G^\alpha = \text{colim}_\alpha \lim_\beta G_{\beta'}^\alpha$$

such that each Y_i stable under G^α and for each i , there exists $B_{\alpha, i}$ that $\mathcal{D}\text{-mod}(Y_i)^{G_{\beta'}^\alpha} = \mathcal{D}\text{-mod}(Y_i)^{G_{\beta'}}$ for all

$\beta' \rightarrow \beta \in B_{\alpha,i}$. Then we can put $\mathcal{D}\text{-mod}(Y_i)^{G^\alpha} = \mathcal{D}\text{-mod}(Y_i)^{G_\beta^\alpha} = \mathcal{D}\text{-mod}(Y_i/G_\beta^\alpha)$. Since G_β^α is smooth of finite type, there is continuous right adjoint $\mathcal{D}\text{-mod}(Y_i) \rightarrow \mathcal{D}\text{-mod}(Y_i)^{G_\beta^\alpha}$, given by $*$ -pushforward along the projection map. Then we have on whole $\mathcal{Y}$, $\mathcal{D}\text{-mod}(\mathcal{Y})^{G^\alpha, \chi} = \lim_\alpha \mathcal{D}\text{-mod}(Y_i)^{G^\alpha, \chi}$. The truncated *averaging* is constructed as limit $\mathbf{Av}_*^{G^\alpha} := \lim_\alpha \mathcal{D}\text{-mod}(\mathcal{Y}) \rightarrow \mathcal{D}\text{-mod}(\mathcal{Y})^{G^\alpha, \chi}$ and finally

$$\mathbf{Av}_*^{\mathcal{G}, \chi} = \lim_\alpha \lim_i \mathbf{Av}_*^{G^\alpha, \chi} : \mathcal{D}\text{-mod}(\mathcal{Y}) \rightarrow \mathcal{D}\text{-mod}(\mathcal{Y})^{\mathcal{G}, \chi}$$

which is right adjoint of $\mathbf{oblv}^{\mathcal{G}, \chi}$, but NOT *continuous* in general. We can regard global Whittaker category as a similar construction applied to $(_{x, \infty} \overline{\mathbf{Bun}}_N)_{\text{good at } \underline{y}}$, in fact, we have continuity in this case.

Proposition 3.15. *The inclusion of category*

$$\text{Whit} \left((_{x, \infty} \overline{\mathbf{Bun}}_N)_{\text{good at } \underline{y}} \right) \hookrightarrow \mathcal{D}\text{-mod} \left((_{x, \infty} \overline{\mathbf{Bun}}_N)_{\text{good at } \underline{y}} \right)$$

admits a continuous right adjoint.

We refer the proof in [Gai21, Corollary 4.6.5]. The next step is to get rid of $\underline{y}$, we need the following property:

Proposition 3.16. *Let $\underline{y} = \underline{y}' \cup \underline{y}''$, then $(_{x, \infty} \overline{\mathbf{Bun}}_N)_{\text{good at } \underline{y}} = (_{x, \infty} \overline{\mathbf{Bun}}_N)_{\text{good at } \underline{y}'} \cap (_{x, \infty} \overline{\mathbf{Bun}}_N)_{\text{good at } \underline{y}''}$. Consider the restriction*

$$\mathcal{D}\text{-mod} \left((_{x, \infty} \overline{\mathbf{Bun}}_N)_{\text{good at } \underline{y}'} \right) \rightarrow \mathcal{D}\text{-mod} \left((_{x, \infty} \overline{\mathbf{Bun}}_N)_{\text{good at } \underline{y}} \right)$$

We have the following commutative diagrams of DG categories:

$$\begin{array}{ccc} \text{Whit} \left((_{x, \infty} \overline{\mathbf{Bun}}_N)_{\text{good at } \underline{y}'} \right) & \longrightarrow & \text{Whit} \left((_{x, \infty} \overline{\mathbf{Bun}}_N)_{\text{good at } \underline{y}} \right) \\ \downarrow & & \downarrow \\ \mathcal{D}\text{-mod} \left((_{x, \infty} \overline{\mathbf{Bun}}_N)_{\text{good at } \underline{y}'} \right) & \longrightarrow & \mathcal{D}\text{-mod} \left((_{x, \infty} \overline{\mathbf{Bun}}_N)_{\text{good at } \underline{y}} \right) \\ \text{Whit} \left((_{x, \infty} \overline{\mathbf{Bun}}_N)_{\text{good at } \underline{y}'} \right) & \longrightarrow & \text{Whit} \left((_{x, \infty} \overline{\mathbf{Bun}}_N)_{\text{good at } \underline{y}} \right) \\ \uparrow & & \uparrow \\ \mathcal{D}\text{-mod} \left((_{x, \infty} \overline{\mathbf{Bun}}_N)_{\text{good at } \underline{y}'} \right) & \longrightarrow & \mathcal{D}\text{-mod} \left((_{x, \infty} \overline{\mathbf{Bun}}_N)_{\text{good at } \underline{y}} \right) \end{array} \quad \text{and}$$

Definition 3.17. Define the **global Whittaker category** $\text{Whit} \left((_{x, \infty} \overline{\mathbf{Bun}}_N)^{G\text{-level}_n, x} \right)$ to be the full subcategory of $\mathcal{D}\text{-mod} \left((_{x, \infty} \overline{\mathbf{Bun}}_N)^{G\text{-level}_n, x} \right)$ containing objects $\mathcal{F}$ such that for any $y \in X - \{x\}$, the restriction of $\mathcal{F}$ onto $(_{x, \infty} \overline{\mathbf{Bun}}_N)^{G\text{-level}_n, x}_{\text{good at } y}$ lands in $\text{Whit} \left((_{x, \infty} \overline{\mathbf{Bun}}_N)^{G\text{-level}_n, x}_{\text{good at } y} \right)$.

Thanks to Proposition 3.16, the property in the above definition holds for any collection of points $\underline{y}$ in $X - \{x\}$, in particular, the restriction functor $\text{Whit} \left((_{x, \infty} \overline{\mathbf{Bun}}_N)^{G\text{-level}_n, x} \right) \rightarrow \text{Whit} \left((_{x, \infty} \overline{\mathbf{Bun}}_N)^{G\text{-level}_n, x}_{\text{good at } \underline{y}} \right)$ is an equivalence for any $\underline{y}$.

3.4 local-to-global Comparison

We note that the construction of the local Whittaker category can be generalized to $\mathcal{Y} := LG/K_n$ by replacing the category $\mathbf{C}$ with the category of twisted $\mathcal{D}$ -modules on $\mathcal{Y}$. To be precise, the stack $\mathcal{Y}$ classifies the data of

$$\left\langle (\mathcal{F}_G, \gamma, \epsilon) \left| \begin{array}{l} \mathcal{F}_G : G\text{-torosr on } X, \epsilon \text{ level structure at } x \\ \gamma : \mathcal{F}_G|_{X-x} \xrightarrow{\cong} \mathcal{F}_{G, \rho(\Omega_X)}^0|_{X-x} \text{ trivialization.} \end{array} \right. \right\rangle$$

where $\mathcal{F}_{G, \rho(\Omega_X)}^0$ is the induced G -torsor from $\rho(\Omega_X)$ (so actually we use $\mathcal{Y}$ as abbreviation for $(LG/K_n)_{\rho(\Omega_X)}$). The above definition does not depend on choice of point $x \in X$. Moreover, we can fix a point $x \in X$ and there is a forgetful map

$$\pi : \mathcal{Y} \rightarrow ({}_{x, \infty} \overline{\text{Bun}}_N)^{G\text{-level}_n, x}$$

that maps $(\mathcal{F}_G, \gamma, \epsilon)$ to $(\mathcal{F}_G, \epsilon)$ with the tautological N -reduction structure (i.e. Plucker embeddings). For fixed x , we can consider $\text{Ran}(X)_x$ that encodes the subsets of points in X that contain x , which can be expressed as $\text{colim}_I X^I \times_X \{x\}$. Accordingly, we can define Ran_x -version of $\mathcal{Y}$, denoted by $\mathcal{Y}_{\text{Ran}_x}$, that classifies the data of

$$\left\langle (\underline{x}, \mathcal{F}_G, \gamma, \epsilon) \left| \begin{array}{l} \underline{x} \in \text{Ran}_x, \mathcal{F}_G : G\text{-torosr on } X, \epsilon \text{ level structure at } x \\ \gamma : \mathcal{F}_G|_{X-\underline{x}} \xrightarrow{\cong} \mathcal{F}_{G, \rho(\Omega_X)}^0|_{X-\underline{x}} \text{ trivialization.} \end{array} \right. \right\rangle$$

We can compare the data of $\mathcal{Y}$ and $\mathcal{Y}_{\text{Ran}_x}$ and observe that there are maps

$$\text{unit}_{\text{Ran}_x} : \text{Ran}_x \times \mathcal{Y} \rightarrow \mathcal{Y}_{\text{Ran}_x}; \quad \pi_{\text{Ran}_x} : \mathcal{Y}_{\text{Ran}_x} \rightarrow ({}_{x, \infty} \overline{\text{Bun}}_N)^{G\text{-level}_n, x}$$

where $\text{unit}_{\text{Ran}_x}$ is a closed embedding. And π can factor as

$$\pi : \mathcal{Y} \rightarrow \text{Ran}_x \times \mathcal{Y} \xrightarrow{\text{unit}_{\text{Ran}_x}} \mathcal{Y}_{\text{Ran}_x} \xrightarrow{\pi_{\text{Ran}_x}} ({}_{x, \infty} \overline{\text{Bun}}_N)^{G\text{-level}_n, x}$$

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